ENCODING SYNTACTIC OBJECTS AND MERGE OPERATIONS IN FUNCTION SPACES

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Abstract. We provide a mathematical argument showing that, given a representation of lexical items as functions (wavelets, for instance) in some function space, it is possible to construct a faithful representation of arbitrary syntactic objects in the same function space. This space can be endowed with a commutative non-associative semiring structure built using the second Rényi entropy. The use of this entropy function can be interpreted as a measure of diversity on the representation of the lexical items. The resulting representation of syntactic objects is compatible with the magma structure. The resulting set of functions is an algebra over an operad, where the operations in the operad model circuits that transform the input wave forms into a combined output that encodes the syntactic structure. A neurocomputational realization of these circuits is completely determined by a realization of a single binary gate that implements a minimization and an entropy computation. The action of Merge on workspaces is faithfully implemented as action on these circuits, through a coproduct and a Hopf algebra Markov chain. The motivation for this mathematical question is to develop possible general theoretical guidelines, based on algebraic properties, that can constrain neurocomputational implementations of Merge and therefore may suggest novel constraints for experimental testing. The results obtained here provide a constructive argument showing the theoretical possibility of a neurocomputational realization of the core computational structure of syntax. We also present a particular case of this general construction where this type of realization of Merge is implemented as a cross frequency phase synchronization on sinusoidal waves. This also shows that Merge can be expressed in terms of the successor function of a semiring, thus clarifying the well known observation of its similarities with the successor function of arithmetic.

1. Introduction

In the context of Biolinguistics, an important challenge is identifying neurocomputational mechanisms that can provide a brain correlate of the fundamental computational structure of syntax in human language. An overview of this problem is presented, for instance, in [65], [21]. In the current model of generative linguistics, the Strong Minimalist Thesis (as developed by Chomsky starting around 2013, see [13], [14]), the core computational structure of syntax is described in terms of the "free symmetric Merge" that is responsible for both structure formation (External Merge) and movement (Internal Merge). A significant advantage of this model is that it decouples this core computational structure from other aspects, such as parametric variation across individual languages, or the interface between syntax and semantics, [6]. Another advantage of this recent formulation is that it admits a mathematical formalization (as developed in [35]). This allows for significant advances in terms of formulating some linguistic questions in a form suitable for the employment of mathematical reasoning and modeling. In this paper, we discuss how the mathematical formulation can be used to gain some theoretical insights on the question of possible neurocomputational realization of the Merge operation.

Different aspects of this broad question on neurocomputational realizations of syntactic structures have been analyzed, see for instance [4], [24], [25], [41], [53]. (Additional literature will be

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discussed briefly in §7.) In this paper we take a more theoretical viewpoint and we give a mathematical proof that, if lexical items are encoded as brain waves (which mathematically means that there is an embedding of the set of lexical items in a space of wavelet functions), then this map can be extended to an embedding of syntactic objects, which moreover preserves the algebraic structure (non-associative commutative magma) that is crucial to structure formation. We show that this embedding can be obtained through neural circuits made of a composition of gates that realize a simple binary operation consisting of a minimization and an entropy computation. It is important to stress that this argument is a theoretical (constructive) proof of concept: it does not imply that an actual neurocomputational realization will necessarily have to be exactly of this form, but it shows a possible theoretical mechanism through which such a realization (with the correct computational properties) can be obtained.

Thus, we focus here on the problem of how to create an embedding of syntactic objects, and of the action of Merge on workspaces, inside a space of functions, in terms of a (nonlinear) superposition of certain wavelets (elements of some function space). The reason for investigating this type of encoding as functions (typically some kind of wavelets) comes from neurolinguistic questions.

It was variously observed in the neuroscience of language that speech and language parsing and processing involves measurable effects of phase synchronization, phase coupling, phase-amplitude coupling (PAC) in brain waves and oscillations, for instance [2], [3], [10], [15], [16], [19], [34], [41] [48], [50], [51], [58], [60], [62], [63], [66], [68], [74], [75]. An important question is how to separate effects due to statistical properties of strings of text (similar to mechanisms now seen in artificial models of language production) from effects that capture the core computational mechanism of syntax and the hierarchical structures present in language. Most of the literature in the neuroscience of language focuses on effects of speech listening. There are a number of empirical results that point more specifically to connections between synchronization and syntax, for instance [1], [10], [49], [73]. In [73] in particular, the statistical and the structural effects and their interactions are analyzed and distinguished. Ultimately, what one would like to achieve is a theoretical model that can be fully formalized in a mathematical and computational way, and that leads to empirically verifiable descriptions of how the structure of syntax and the fundamental structure building operations of syntax, as a computational system, can be realized in neural population activity and in observable rhythms and brain oscillations.

While we do not try here to address this complex problem, we will provide a flexible template of how an operation with the prescribed algebraic properties of Merge and the resulting syntactic objects generated by a magma operation can be faithfully embodied in spaces of wavelet functions. This mathematical model can in principle be flexibly adapted to at least some of the existing theoretical models proposed in language neuroscience, providing a way of incorporating an operation with the correct formal properties of the syntactic Merge.

For instance, we discuss briefly some aspects of two recent theoretical models: the Compositional Neural Architecture model of [42] (see also [43], [44], [45]), based on a symbolic-connectionist DORA (Discovery of Relations by Analogy) model and on a “time-based binding” that produces synchronization phenomena related to hierarchical structures, and the ROSE (Representation, Operation, Structure, Encoding) theoretical model of [53], [54], [55], that proposes a neurocomputational realization of Merge in terms of brain oscillations and synchronization and modulation phenomena across different types of brain waves. From these and other discussions in neurolinguistics on representations of hierarchical structures starting from waves or time series representations of lexical items, we develop a special form of our model where the action of Merge in such representations is implemented via phase synchronizations.

Another motivation comes from brain-machine interface experiments that indicate neural representations of internal and vocalized words, at both the single neuron and population levels, [18], [47], [52], [72]. Current brain-machine interface models are typically based on phonetic (not lexical or semantic) representations, and are use ML analysis of high gamma activity for word prediction, rather than identifying interpretable neural representations. Further experiments (see [61]) use cross-frequency coupling for decoding, rather than just high or low frequency activity. All these different experiments do, in any case, raise the theoretical question of whether representations of hierarchical structures of syntax would also be similarly possible, going beyond just speech as a sequence of words, and on to the level of language as hierarchical structures, [31], [32]. We do not discuss in this paper the interface between syntax and morphology (which can be articulated within the mathematical formulation of Minimalism, [67]), though brain waves models of the syntax-morphology interface have also been proposed in [42], [53]: an investigation of possible neurocomputational models of the algebraic structure described in [67] is an interesting question.

The point of view we follow in this note is simply to consider the underlying mathematical question, and provide a constructive proof that it is possible indeed to faithfully and efficiently encode syntactic objects with the algebraic structure of commutative non-associative magma, and movement by Internal Merge, starting with a given embedding of the set of lexical items and syntactic features in a function space.

As a model of the core computational structure of syntax, we follow here the formulation of Chomsky's free symmetric Merge in the Strong Minimalist Thesis, [13], [14], in the mathematical form developed in the work of Marcolli, Chomsky, Berwick, [35].

The construction that we present here can be seen in the same light as the model of syntax-semantics interface discussed in [35]. However, we focus here on a somewhat different type of property and a very different interpretation. At the level of formal properties, the embedding we consider takes values in a semiring, where the non-associative but commutative semiring addition will account for the magma structure operation on syntactic objects. The action of Merge can then be described as an action on circuits that compute this embedding of syntactic objects starting from the lexical items. The construction we present crucially uses an information measure, to obtain the addition operation in the semiring, which can be seen as a diversity measure for the dictionary encoding the lexical items.

This model is simply aimed at showing that such encoding of syntactic objects and syntactic operations is theoretically possible, given an encoding of lexical items by certain wavelet functions: it does not necessarily provide a direct realization in neurocomputational terms. We will give some indication as to the realizability of the entropy computations required in this model, to show that it is not unreasonable to expect those to be implementable in neuronal circuits.

In particular, in the last section of this paper, we discuss some aspects of the existing literature and possible relations to the model we present here. We include a brief discussion of some aspects of the Compositional Neural Architecture model of [42] and of the ROSE model of [54], in the light of the construction presented here. Although we do not try to formalize or implement any of these existing models, we focus on one particular aspect they share, which is the role of cross-frequency and phase-phase synchronization of waves. We use this as motivation for presenting a simpler instantiation of the general model described in the previous sections in terms of phase synchronization of simple sinusoidal waves, where one can make some properties of our model more explicit.

In particular, this construction via phase synchronizations shows an interesting property of Merge, namely how Merge can be expressed in terms of a "successor function". Chomsky had noted, on various occasions in the past, how Merge bears some similarities to the successor function of arithmetic (see for instance [12]). However, these two operations have different algebraic properties, so they cannot be directly related. On the other hand, the thermodynamic semirings of [39] that we use in this paper also have a successor function (generalizing the usual successor function of arithmetic associated to the semiring $\mathbb{N}$), as shown in [39]. We show here that this kind of successor function is indeed the one that can be used to describe Merge. This clarifies in a mathematically precise way, the relation between Merge and the successor function.

Our model of the Merge operation in function spaces of wavelet-like signals also answers a question posed by Andrea Martin in [42]: whether composition via vector addition is a sufficient operation for compositionality in natural language. We confirm the fact that the compositionality of Merge does not have the properties of a tensor product type operation (multiplicative combination), but rather behaves like a sum (additive combination), as observed in [42]. However, we show that vector addition does not have the right algebraic properties to implement structure formation by Merge. On the other hand, the solution we present, that does faithfully encode the properties of Merge, is obtained by optimizing over convex combinations of vectors using an entropy functional that can be chosen in a natural way (the Rényi entropy) so as to obtain the correct algebraic properties for the Merge operation.

2. Syntactic Objects and Weighted Combinations

We first analyze the magma of syntactic objects and how it can be encoded in a space of functions. As in [35], we denote by $\mathcal{SO}_0$ the (finite) set of lexical items and syntactic features, and by $\mathcal{SO}$ the (countable) set of syntactic objects. The set $\mathcal{SO}$ has an algebraic structure: it is the non-associative, commutative magma generated by the set $\mathcal{SO}_0$,

$$\mathcal{SO} = \mathrm{Magma}_{na,c}(\mathcal{SO}_0, \mathfrak{M})$$

where $\mathfrak{M}$ denotes the magma operation. This implies that $\mathcal{SO}$ can be identified with the set $\mathfrak{T}_{\mathcal{SO}_0}$ of abstract (no-planar) binary rooted trees with leaves decorated by elements of $\mathcal{SO}_0$. As in [35], we will use the notations $\mathcal{SO}$ and $\mathfrak{T}_{\mathcal{SO}_0}$ interchangeably.

Let $\mathcal{F}$ denote a function space. We want to be fairly broad for now in terms of the properties of this space, but it would typically be an infinite dimensional topological vector space (or Banach space, or Hilbert space) of functions on some underlying topological space X. We think of the space X as representing areas of the brain as well as an interval of time over which observations of brain activity are performed. We can assume that X is a compact space. This will include cases where $\mathcal{F}$ is the space of continuous functions $\mathcal{C}(X, \mathbb{R})$, or of smooth functions $\mathcal{C}^\infty(X, \mathbb{R})$, or integrable or square-integrable functions, $L^1(X, \mathbb{R})$ or $L^2(X, \mathbb{R})$. We can also allow generalizations that include spaces of distributions, rather than functions. In practice, while what we say in this section will hold at this level of generality, we can for simplicity think of the case of continuous functions on an interval $X = [0, 1]$, which suffices to describe functions of time, over a certain window of observation (though in general we would want, more realistically to include in X also positional coordinates identifying brain locations).

Of particular interest is spaces of functions generated by a wavelet basis, as wavelet analysis is ordinarily employed for signal analysis and denoising, by decomposing a signal in a wavelet basis. For example, in wavelet analysis of magnetoencephalography (MEG) data (see [18]) a wavelet decomposition using discrete wavelet transform provides components corresponding to the high-gamma (62–125 Hz), gamma (31–58 Hz), beta (16–30 Hz), alpha (8–16 Hz), and theta (4–8 Hz) frequency bands of the neural signal, with the higher frequency components removed for denoising. In [18], the spectral-temporal characteristics of the signal are encoded in a representation, obtained via continuous wavelet transform, of the denoised signal in a basis of Morlet wavelets.

Let then $\mathcal{F}$ be a linear space of functions endowed with a chosen basis as above. Suppose given an injective function $\varphi : \mathcal{SO}_0 \to \mathcal{F}$. This would represent an encoding of lexical items (together with syntactic features) as certain linear combinations in that basis (say, combinations of wavelets). Such a map can be constructed empirically by methods such as those discussed in [18], [72]. We will just assume here that it is a given input of our theoretical model.

There is an important issue here that concerns the difference between words and lexical items and syntactic features. While empirically assigning a wavelet representation to words is done in neuroscience experiments, an analogous encoding for the more abstract elements of the set $\mathcal{SO}_0$ is a more delicate issue. Since we are primarily interested here in outlining the mathematical structure of possible neurocomputational syntax encoding models, we will for simplicity ignore these differences at this stage. (Linguistic proposals for a simplifying setting for $\mathcal{SO}_0$ can be found, for instance, in [20]. We do not need to discuss this question here, as we only take the existence of a representation $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ as a given input for our construction.)

Our main working assumptions about the function $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ are summarized as follows.

Remark 2.1. We assume that the map $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ is injective, namely that its image can disambiguate any different items in $\mathcal{SO}_0$. We also assume the stronger property that, with respect to the vector space structure of $\mathcal{F}$, all the $\varphi(\alpha)$, for $\alpha \in \mathcal{SO}_0$, are linearly independent vectors.

The question we consider here is how to extend the map $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ to a map $\varphi : \mathcal{SO} \to \mathcal{F}$ that is still an embedding, and that also encodes the magma structure of $\mathcal{SO}$. We also want this map to be in some sense a solution to an optimization problem in the computational and information theoretic sense. The idea that the Merge operation in syntax should be characterized by optimization properties in terms of efficiency of computation (often referred to in the literature as "third factor" constraints) has been a key idea in the development of Minimalism, especially in its current form. The mathematical model of Minimalism of [35] formulates some of these notions of optimality in a more directly quantifiable way, which in turn can be used for deriving linguistic consequences, as in [38]. Here we will identify another possible form of optimality, in the specific realization that we propose in function spaces, namely the fact that the nonassociativity property of Merge is obtained through an entropy minimization, and the particular choice of entropy functional used for minimization (the second Rényi entropy) has itself optimality properties, as we discuss in §3.2.1.

The key parts of this problem can be rephrased as the following questions.

Question 2.2.
(1) Given an abstract binary rooted tree $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with $\{\alpha_\ell\}_{\ell \in L(T)}$ the elements $\alpha_\ell \in \mathcal{SO}_0$ assigned to the leaves $\ell \in L(T)$, construct a (not necessarily linear) combination of the functions $\varphi(\alpha_\ell) \in \mathcal{F}$ that faithfully encodes the combinatorial tree structure of T.
(2) Show that this construction is compatible with the magma structure, in the sense that, if $\varphi(T)$ and $\varphi(T')$ are obtained as above, then $\varphi(\mathfrak{M}(T, T')) = \Gamma(\varphi(T), \varphi(T'))$ for a commutative non-associative binary operation Γ on $\mathcal{F}$.
(3) Show that the function computing $\varphi(T)$ from the $\varphi(\alpha_\ell)$ is optimal with respect to an entropy/information functional.

We will answer these three questions in steps. We first show in §2.1 that it is possible to associate to a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with a head function (syntactic head) a probability distribution $A = (a_\ell)_{\ell \in L(T)}$ on the set of leaves $L(T)$, so that $a_\ell = P_\ell(\Lambda)$ is a polynomial function of variables $\Lambda = (\lambda_v)$ with $\lambda_v \in [0, 1]$ associated to each non-leaf vertex $v \in V^o(T)$. We show that the function $A = P(\Lambda)$ completely determines the pair (T, h_T) of the syntactic object and the head function. This construction gives an embedding of syntactic objects into a much larger space of functions than the one where the lexical items are embedded, as the function associated to T depends on a collection $\Lambda = (\lambda_v)$ of auxiliary variables, in addition to the variables in X that the functions $\varphi(\alpha_\ell)$ depend on. This is not yet satisfactory because one wants both lexical items and syntactic objects to all live in the same function space (without a rapidly growing set of additional variables) and, moreover, one wants the images of syntactic objects in this vector space to still carry the algebraic operation that gives the magma structure and that is the base of the structure formation via Merge. To move towards that goal, we then observe in §2.3 that the operation $\Gamma(\varphi(T), \varphi(T')) = \lambda\varphi(T)+(1-\lambda)\varphi(T')$ that combines two functions in a convex combination has some of the algebraic properties we want for Merge (non-associativity) but not others (commutativity). This suggests that a modification of this kind of operation that achieves commutativity but maintains non-associativity, and that also identifies an optimal choice of the value of λ, can provide the next step of the construction. We do this in §3 by endowing the space of functions $\mathcal{F}$ with a non-associative commutative addition that is obtained as a deformation of the $(\min, +)$ semiring (tropical semiring) with a deformation parameter that behaves like a thermodynamical inverse temperature and where the convex combination $\Gamma(\varphi(T), \varphi(T')) = \lambda\varphi(T) + (1-\lambda)\varphi(T')$ is corrected by adding a term $-\beta^{-1}S(\lambda, 1-\lambda)$, where S is an information measure. The resulting $\Gamma(\varphi(T), \varphi(T'))-\beta^{-1}S(\lambda, 1-\lambda)$ is then minimized over λ: for suitable choices of S this operation is indeed non-associative but commutative. The question then is whether with this operation one can obtain embeddings of syntactic objects in $\mathcal{F}$. This question is answered in §4.2 and §4.3, with some further discussion of the remaining cases in §4.4. The compatibility with the magma operation is discussed in §5 and the implementation of the Hopf algebra structure and the Merge operation in §6.

2.1. **Polynomial functions from trees.** As in [35], a head function h_T on a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ is a function $h_T : V^o(T) \to L(T)$ from the non-leaf vertices of T to the leaves, with the property that $h_T(v) \in L(T_v)$, where $T_v \subset T$ is the accessible term at v (the full subtree rooted at v) and that if $T_w \subset T_v$ and $h_T(v) \in L(T_w)$ then $h_T(v) = h_T(w)$. This abstract property is designed to capture the main properties of the syntactic head and (as shown in [35]) it agrees with the definition of head given in [11]. As shown in [35], assigning a head function h_T to T is the same thing as assigning at each vertex $v \in V^o(T)$ a marking to one of the two edges below v (the direction pointing to the leaf $h_T(v)$). In particular, in a pair $(T, h_T(v))$ there is an assigned way of selecting one of the two edges below each vertex. The presence of a syntactic head is part of the requirements of a well formed syntactic structure. We write $h : T \mapsto h_T$ for an assignment of a head function and $\mathrm{Dom}(h) \subset \mathcal{SO}$ for the set of syntactic objects that have a head function. As discussed in [35] and [37], $\mathrm{Dom}(h)$ is not a submagma: incorporating head functions leads to extending the magma structure to a hypermagma. As we will see, this does not matter in the end in our argument, since the dependence on the head function will be eliminated at a later step of our construction. In other words, the realization of Merge that we propose does not in itself depend on the use of the head function. (Indeed this has to be the case, as head functions are not compatible with the magma structure, as discussed in §1.13 of [35] and in [37].) On the other hand, a head function is needed for well formed phases and for interpretation at the syntax-semantics interface. We will discuss in §7.8 how the head function can be modeled, in a particular form of the model we discuss in this paper.

We first extend the map $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ to a map $\varphi : \mathrm{Dom}(h) \subset \mathcal{SO} \to \mathcal{F} \otimes_{\mathbb{R}} \mathbb{R}[\underline{\Lambda}]$ with values in the tensor product of the vector space $\mathcal{F}$ with polynomials in a countable set of variables $\underline{\Lambda}$.

Proposition 2.3. *Given an injective map $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ whose image consists of a collection of linearly independent vectors in $\mathcal{F}$, and a syntactic object $T \in \mathcal{SO} = \mathfrak{T}_{\mathcal{SO}_0}$ that is in* $\mathrm{Dom}(h)$*, let $V^o(T)$ denote the set of non-leaf vertices of T. Let $\varphi(\alpha_\ell)$ be the functions associated to the elements $\alpha_\ell \in \mathcal{SO}_0$ at the leaves $\ell \in L(T)$. These determine a function $\varphi(T) \in \mathcal{F}[\{\lambda_v\}_{v\in V^o(T)}$,* *the vector space of polynomials in a set of variables $\Lambda = \{\lambda_v\}_{v\in V^o(T)}$ associated to the non-leaf vertices,*

$$\varphi(T)(\Lambda) = \sum_{\ell\in L(T)} a^T_\ell(\Lambda)\varphi(\alpha_\ell)\,, \tag{2.1}$$

where the $a^T_\ell(\Lambda)$ are polynomials in the variables λ_v. The polynomials $a^T_\ell(\Lambda)$ are determined recursively via the following relation. Let $T = \mathfrak{M}(T_1, T_2) \in \mathrm{Dom}(h)$ with root vertex v_T and with $h_T(v_T) = h_{T_1}(v_{T_1})$, where v_{T_1} is the root vertex of T_1. Then

$$a^T_\ell = \begin{cases} \lambda_{v_T}\, a^{T_1}_\ell & \textit{for } \ell \in L(T_1) \subset L(T)\,, \\ (1-\lambda_{v_T})\, a^{T_2}_\ell & \textit{for } \ell \in L(T_2) \subset L(T)\,, \end{cases} \tag{2.2}$$

where the recursion starts at $T = \ell$ a single leaf labelled by α_ℓ, for which we take $a^\ell_\ell = 1$. The case of $h_T(v_T) = h_{T_2}(v_{T_2})$ is analogous with λ_v replaced by $1 - \lambda_v$.

Proof. The construction of $\varphi(T)$ follows the construction of T in the magma $\mathcal{SO}$, starting with the elements of $\mathcal{SO}_0$. Given a pair $\alpha, \beta \in \mathcal{SO}_0$ and a cherry tree substructure of T of the form

$$T_v = \{\alpha, \beta\} = \widehat{\alpha\ \ \beta}\,,$$

with root vertex v, the functions $\varphi(\alpha)$ and $\varphi(\beta)$, and a variable λ_v, form the combination

$$\varphi(T) = \lambda\varphi(\alpha) + (1-\lambda)\varphi(\beta)\,,$$

if α is the head of T (and with λ and $1-\lambda$ exchanged otherwise). When further applying the magma operation $\mathfrak{M}$ we assign to

$$\mathfrak{M}(T, \gamma) = \{\{\alpha, \beta\}, \gamma\} = \widehat{\widehat{\alpha\ \ \beta}\ \ \gamma}$$

(where the trees are non-planar) the combination

$$\lambda'\lambda\varphi(\alpha) + \lambda'(1-\lambda)\varphi(\beta) + (1-\lambda')\varphi(\gamma)\,,$$

where λ, λ' are two parameters. The combination is depicted here in the case where α is the head of the resulting tree. We see that this procedure can be continued, resulting in an expression $\varphi(T)$ that depends on a collection of parameters $\{\lambda_v\}_{v\in V^o(T)}$, with $V^o(T)$ the set of non-leaf vertices of T. This can be seen inductively as phrased in the statement. If we have constructed $\varphi(T)(\Lambda)$ and $\varphi(T')(\Lambda')$, where $\Lambda = (\lambda_v)_{v\in V^o(T)}$ and $\Lambda' = (\lambda'_w)_{w\in V^o(T')}$, for two given syntactic objects $T, T' \in \mathcal{SO}$, then we define

$$\varphi(\mathfrak{M}(T, T'))(\Lambda, \Lambda', \lambda_u) = \lambda_u\varphi(T)(\Lambda) + (1-\lambda_u)\varphi(T')(\Lambda')\,,$$

where λ_u is the additional parameter associated to the new root vertex of $\mathfrak{M}(T, T')$, and we assume that the head of $\mathfrak{M}(T, T'$ is the head of T (if it is the head of T' we replace $\lambda_u \leftrightarrow 1-\lambda_u$). The result of this construction can be seen as a function

$$\varphi(T) = \sum_{\ell\in L(T)} a_\ell\, \varphi(\alpha_\ell)\,,$$

where $\alpha_\ell \in \mathcal{SO}_0$ are the items associated to the leaves $\ell \in L(T)$, with $\varphi(\alpha_\ell) \in \mathcal{F}$, where the coefficients a_ℓ are polynomial functions of the variables λ_v, for $v \in V^o(T)$,

$$a_\ell = P_\ell(\lambda_v) \in \mathbb{Z}[\{\lambda_v\}_{v\in V^o(T)}] \subset \mathbb{R}[\{\lambda_v\}_{v\in V^o(T)}]$$

as stated. Since the parameters λ_v are regarded as variables, and not evaluated here at a particular point of the unit interval, the resulting $\varphi(T)$ is an element of the vector space of polynomials in the variables λ_v, with coefficients in the vector space $\mathcal{F}$, namely, in $\mathcal{F} \otimes \mathbb{R}[\{\lambda_v\}_{v\in V^o(T)}]$ (which is a vector space not a ring, unless $\mathcal{F}$ is also an algebra and not just a vector space). $\square$

The presence of the head function h_T on T is simply used here to decide which of the edges below a vertex v contributes a factor λ_v (marked edge in the head direction) or $1-\lambda_v$ (the other edge). More precisely, this can be phrased as the following observation.

Remark 2.4. The construction of $\varphi(T)$ follows the generative process of the magma $\mathcal{SO}$, since the recursive construction uses the magma operation $T = \mathfrak{M}(T_1, T_2)$. However, in the assignment

$$\varphi(T) = \lambda\varphi(T_1) + (1-\lambda)\varphi(T_2)\,,$$

the presence of a head function on T is used to decide which of the two terms gets the variable λ and which $1-\lambda$. The domain $\mathrm{Dom}(h)$ is not a submagma of $\mathcal{SO}$: in general $\mathfrak{M}(T_1, T_2)$ is not necessarily in $\mathrm{Dom}(h)$ even though T_1 and T_2 are. Thus, in Proposition 2.3 we need to assume $T \in \mathrm{Dom}(h)$.

Corollary 2.5. *A head function h_T on a syntactic object T determines a partition of the set of vertices $V(T)$ into a disjoint union of paths γ_ℓ for $\ell \in L(T)$, whose edges are the marked edges determined by the head function. The recursive form* (2.2) *gives the explicit form of the polynomial dependence $A = P(\Lambda)$ as*

$$a^T_\ell = \prod_{v\in\gamma_{v_0,\ell}} \epsilon_v\,, \quad \textit{with } \epsilon_v = \left\{ \begin{array}{ll} \lambda_v & e_v \in \gamma_{h_T(v)} \\ 1-\lambda_v & e_v \notin \gamma_{h_T(v)}\,, \end{array} \right. \tag{2.3}$$

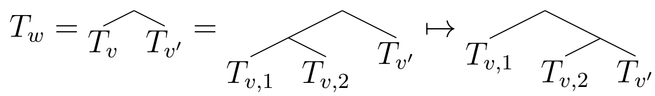
with v_0 the root of T, where $\gamma_{v_0,\ell}$ is the path from v_0 to the leaf ℓ and e_v denotes the edge of the path $\gamma_{v_0,\ell}$ with source vertex v (in the orientation away from the root).

Proof. As recalled above, the head function marks an edge below each vertex. The paths γ_ℓ can be obtained by following for each $\ell \in L(T)$ the path along these marked edges Some of the paths γ_ℓ are trivial and consist of a single leaf and no edges. For the nontrivial path, the highest vertex v_ℓ on the path is the maximal projection of the head ℓ, see [35] for a discussion of all these properties of the head function. By (2.2) , at each step along the path $\gamma_{v_0,\ell}$ from the root v_0 to the leaf ℓ, we multiply by either a factor λ_v is the edge is marked or $1-\lambda_v$ if it is not, that is, according to whether the edge e_v is on the path $\gamma_{h_T(v)}$ (the path from v to the leaf $h_T(v)$) or not, resulting in the product (2.3). ☐

2.2. **Reconstructing the tree.** We show the injectivity of the map constructed in Proposition 2.3, namely the fact that the syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$ can be reconstructed from its image $\phi(T)$ in $\mathcal{F} \otimes \mathbb{R}[\Lambda]$.

Proposition 2.6. *One can reconstruct the tree T and the head function h_T from the function $\varphi(T)$ of the variables λ_v, under the assumption that the $\varphi(\alpha)$, for $\alpha \in \mathcal{SO}_0$, are linearly independent.*

Proof. For a fixed set of leaves L and distinct items $\alpha_\ell \in \mathcal{SO}_0$, for $\ell \in L$, associated to the leaves, we need to show that, if $T \neq T'$ are two non-isomorphic trees with $L(T) = L(T') = L$, then for some $\ell \in L$ the polynomials $a_\ell(T)$ and $a_\ell(T')$ are also different, $a_\ell(T) \neq a_\ell(T')$, so that $\varphi(T) \neq \varphi(T')$. First observe that we can associate to the pair (T, Λ) with $\Lambda = (\lambda_v)_{v\in V^o(T)}$ a point in the BHV moduli space (see [5] and §3.4.2 of [35]), by endowing each edge e of T with a weight w_e equal to either λ_v or $1-\lambda_v$, depending on whether in the function $\varphi(T)$ the polynomials a^T_ℓ of all the leaves ℓ in the subtree below the edge e contain a factor λ_v or $1-\lambda_v$. In the BHV moduli space a non-planar binary rooted tree T with assigned weights $w_e > 0$ at the edges corresponds to a point inside one of the open cells of the moduli space. The walls between the open cells correspond to degenerate trees with higher valence vertices, where one (or more in the deeper strata) of the edge weights take the limiting value $w_e = 0$. Thus, passing from one to another non-isomorphic tree with the same labelled leaves is achieved by a number of wall crossings in the BHV moduli space. Each such wall crossing corresponds to one edge e with source v moving across the vertex w above its source v, by contracting to zero length the edge between v and w and creating a non-zero length edge on the other branch below w. With v' denoting the other vertex below w, and $T_v = \mathfrak{M}(T_{v,1}, T_{v,2})$, each such crossing corresponds to a change



which corresponds to a change

$$\lambda'\lambda\varphi(T_{v,1}) + \lambda'(1-\lambda)\varphi(T_{v,2}) + (1-\lambda')\varphi(T_{v'}) \mapsto$$
$$\lambda'\varphi(T_{v,1}) + (1-\lambda')\lambda\varphi(T_{v,2}) + (1-\lambda')(1-\lambda)\varphi(T_{v'})$$

in the associated function.

The functions $\varphi(T_{v,1})$, $\varphi(T_{v,2})$, $\varphi(T_{v'})$ are linearly independent in $\mathcal{F}$ (for any fixed values of the parameters λ_v) as linear combinations of different sets of α_ℓ (since the sets $L(T_{v,1})$, $L(T_{v,2})$ and $L(T_{v'})$ do not overlap). Thus, by the same argument given above, the two expressions

$$\lambda'\lambda\varphi(T_{v,1}) + \lambda'(1-\lambda)\varphi(T_{v,2}) + (1-\lambda')\varphi(T_{v'})$$
$$\lambda'\varphi(T_{v,1}) + (1-\lambda')\lambda\varphi(T_{v,2}) + (1-\lambda')(1-\lambda)\varphi(T_{v'})$$

are different for any value $\lambda, \lambda' \notin \{0,1\}$. Since each successive wall crossing will involve similar transformations at a different internal vertices w, affecting different subsets of leaves, each time we obtain different resulting functions, so that the different trees T with same $L = L(T)$ will give rise to different functions $\varphi(T)$. For trees in $\mathcal{SO}$ that do not have the same set of leaves the resulting functions are necessarily different because they either depend on a different set of variables, if the set of leaves do not have the same cardinality, or have different elements in $\mathcal{SO}_0$ at the leaves hence give combinations of different sets of independent vectors. This means that, in all cases, the functions $\varphi(T)$ suffice to reconstruct the syntactic objects $T \in \mathcal{SO}$. Corollary 2.5 shows that the head function h_T is also encoded in $\varphi(T)$: indeed, assigning h_T is the same thing as assigning which edge below each vertex is marked, and this can be obtained from the formula (2.3), by checking which factor is a λ_v or a $1-\lambda_v$. ☐

Combining Proposition 2.3 and Proposition 2.6 we obtain the following result.

Corollary 2.7. *Let $\underline{\Lambda} = \cup_{T\in\mathfrak{T}}\{\lambda_{T,v} \,|\, v \in V^o(T)\}$ be a countable set of variables with $T \in \mathfrak{T}$ ranging over the countable set of all non-planar binary rooted trees. The construction of Proposition 2.3 extends an injective map $\varphi : \mathcal{SO}_0 \to \mathcal{F}$, whose image consists of a collection of linearly independent vectors in $\mathcal{F}$, to an injective map*

$$\varphi : \mathrm{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0} \to \mathcal{F} \otimes_{\mathbb{R}} \mathbb{R}[\underline{\Lambda}]\,. \tag{2.4}$$

2.3. **Combination operations.** For any evaluation of the parameters λ_v at particular points $\lambda_v \in (0,1)$, we obtain a function $\varphi(T) \in \mathcal{F}$. In general we do not expect that fixing the values of the λ_v variables will still yield an injective map of the syntactic objects to $\mathcal{F}$. Thus, the question, in terms of these evaluations of $\varphi(T)(\Lambda)$ is whether there is an optimal way of choosing the values of the λ_v variables. We discuss in §3 a way to pass from functions $\varphi(T) \in \mathcal{F} \otimes \mathbb{R}[\underline{\Lambda}]$ to functions in $\mathcal{F}$ that is motivated by the algebraic properties of the iterated convex combinations involved in the construction of $\varphi(T)$. We first observe the following property.

Lemma 2.8. *For given $\lambda, \lambda' \in (0,1)$, the operation*

$$\Gamma(\varphi(T), \varphi(T')) = \lambda\varphi(T) + (1-\lambda)\varphi(T') \tag{2.5}$$

is non-associative. It is also non-commutative, except when $\lambda = \lambda' = 1/2$.

Proof. The non-associativity is seen by the fact that, since the $\varphi(\alpha)$, for $\alpha \in \mathcal{SO}_0$, are linearly independent, setting

$$\lambda'\lambda\varphi(\alpha) + \lambda'(1-\lambda)\varphi(\beta) + (1-\lambda')\varphi(\gamma) = \lambda'\varphi(\alpha) + (1-\lambda')\lambda\varphi(\beta) + (1-\lambda')(1-\lambda)\varphi(\gamma)$$

would imply $\lambda'\lambda = \lambda'$ and $\lambda'(1-\lambda) = (1-\lambda')\lambda$ and $1-\lambda' = (1-\lambda')(1-\lambda)$, which would give $\lambda, \lambda' \in \{0,1\}$ which are excluded values. Similarly, by linear independence, we see that $\lambda\varphi(T) + (1-\lambda)\varphi(T') = \lambda\varphi(T') + (1-\lambda)\varphi(T)$ iff $\lambda = 1-\lambda$ and $\lambda' = 1-\lambda'$, i.e. $\lambda = \lambda' = 1/2$. In other words, the lack of commutativity is measured by the transformation that exchanges $\lambda \leftrightarrow (1-\lambda)$. □

We want to improve on this representation, by obtaining a faithful mapping of syntactic objects to a space of functions, so that the resulting functions $\varphi(T)$ are obtained (not necessarily as linear combinations) from the functions $\varphi(\alpha_\ell)$ associated to the lexical items at the leaves, in such a way that the mapping preserves the structure of *commutative, non-associative magma*. In order to do this, we use the formalism of thermodynamic semirings introduced in [39].

3. Thermodynamic Semirings and the Rényi Entropy

We have obtained a faithful encoding of the syntactic objects (with a head function) in a vector space $\mathcal{F} \otimes_{\mathbb{R}} \mathbb{R}[\underline{\Lambda}]$ of functions. The question then is how to improve on this construction so that a faithful encoding can happen in the same space $\mathcal{F}$ and in a way that also encodes the magma operation.

One issue about encoding the magma operation, as already mentioned in Remark 2.4, is that the domain $\mathrm{Dom}(h)$ of a head function is not compatible with the magma structure. This issue can be resolved by passing to hypermagmas as in [37]. There is, however, another issue discussed in Lemma 2.8: commutativity for the operation $\Gamma(\varphi(T), \varphi(T'))$ only holds up to the transformation $\lambda \leftrightarrow 1-\lambda$, while the magma operation is commutative, $\mathfrak{M}(T,T') = \mathfrak{M}(T',T)$.

We show that these two problems (maintaining the same space $\mathcal{F}$ of the embedding and encoding the magma structure) can be solved at the same time using a construction developed in [39] of "thermodynamic semirings" involving an entropy measure. The entropy functional will provide an addition operation that is non-associative and commutative, which will implement the magma operation of syntactic objects, while performing an optimization over the variables λ_v, which reduces the embedding space back to $\mathcal{F}$ by fixing an optimal value of these additional variables.

3.1. **Probabilities and entropy.**

The probability distribution we consider here, associated to a syntactic object T, is the $A = (a_\ell)_{\ell \in L(T)}$ constructed as in Corollary 2.5. We verify that it is indeed a probability distribution as follows.

Lemma 3.1. *For all $T \in \mathfrak{T}_{\mathcal{SO}_0}$ and for all $\Lambda = (\lambda_v)_{v \in V^0(T)} \in (0,1)^{\#V^0(T)}$, the vector $A = (a^T)_{\ell \in L(T)}$, with $a^T_\ell(\Lambda)$ as in Proposition 2.3 is a probability distribution on the set $L(T)$, namely $a^T_\ell(\Lambda) \geq 0$ and*

$$\sum_{\ell \in L(T)} a^T_\ell(\Lambda) = 1\,. \tag{3.1}$$

Proof. It is clear that, if all the λ_v are in $(0,1)$ then the polynomials $a^T_\ell(\Lambda)$ take non-negative values, since each $a^T_\ell(\Lambda)$ is a product $a^T_\ell(\Lambda) = \prod_{w \in \pi_\ell} c_w$ where π_ℓ is the path of vertices from the leaf ℓ to the root vertex of T and each c_w is either λ_w or $1-\lambda_w$, so $a^T_\ell(\Lambda) > 0$. We can also show inductively that $\sum_\ell a_\ell = 1$. For $\#L(T) = 2$ we simply have λ and $1-\lambda$ as the values of a^T_ℓ. Similarly, with the a^T_ℓ equal to $\lambda'\lambda$, $\lambda'(1-\lambda)$, and $(1-\lambda')$ in the case of a tree T with three leaves. Assuming the normalization (3.1) holds up to a given size, it suffices to show it also holds for $\mathfrak{M}(T,T')$, to inductively increase the size. For $\ell \in L(T)$ we have $\sum_\ell a_\ell^T = 1$ and for $\ell' \in L(T')$ we also have $\sum_{\ell'} a_{\ell'}^{T'} = 1$. Then $L(\mathfrak{M}(T,T')) = L(T) \sqcup L(T')$, with $a_\ell^{\mathfrak{M}(T,T')} = \lambda a_\ell^T$ and $a_{\ell'}^{\mathfrak{M}(T,T')} = (1-\lambda) a_{\ell'}(T')$, so that one still has a probability distribution. □

Since $A(T) = (a_\ell(T))_{\ell \in L(T)}$ is a probability distribution, we can compute its entropy/information according to different kinds of information functionals, such as Shannon entropy, Rényi and Tsallis entropies.

Definition 3.2. *A* binary information measure *is a continuous concave function $S : [0,1] \to \mathbb{R}_{\geq 0}$ with $S(0) = 0 = S(1)$ and $S(p) = S(1-p)$. The particular case of the Shannon entropy also satisfies the condition*

$$S(p) + (1-p) S(\frac{q}{1-p}) = S(p+q) + (p+q) S(\frac{p}{p+q}) .$$

We will equivalently write either $S(p)$ or $S(p, 1-p)$, where the latter notation explains the term "binary information measure".

Definition 3.3. *The tropical semiring consists of the set $\mathbb{R} \cup \{\infty\}$ endowed with two operations $\oplus$ and $\odot$. The product operation $\odot = +$ is the ordinary sum of real numbers, extended with the rule that $x + \infty = \infty = \infty + x$ for any $x \in \mathbb{R} \cup \{\infty\}$. The identity element for $\odot$ is 0. The sum operation $\oplus = \min$ is the minimum $x \oplus y = \min\{x, y\}$, with identity element ∞. Both operations satisfy associativity and commutativity and have identity elements, and the product $\odot$ distributes over the sum $\oplus$. However, since there are in general no additive inverses, this is a semiring rather than a ring.*

The following deformations of the tropical semirings were studied in [39]. We summarize here the main construction.

Remark 3.4. A binary information functional S determines a modified addition operation $\oplus_{S,\beta}$ on the same set $\mathbb{R} \cup \{\infty\}$. Instead of $x \oplus y = \min\{x, y\}$, one defines

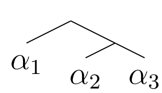
$$x \oplus_{S,\beta} y = \min_{p \in [0,1]} \{ p x + (1-p) y - \beta^{-1} S(p) \} . \tag{3.2}$$

Then $\mathbb{T}_{S,\beta} = (\mathbb{R} \cup \{\infty\}, \oplus_{S,\beta}, \odot)$ is still a semiring (called the *thermodynamic semiring* with entropy S (see [39]). The commutativity of the operation $\oplus_{S,\beta}$ follows from the symmetric property $S(p) = S(1-p)$ of the binary information functional, the fact that 0 is still the additive unit follows from the property $S(0) = 0 = S(1)$. However, in general $\mathbb{T}_{S,\beta}$ is *non-associative*. Associativity in fact holds if and only if S is the Shannon entropy. For any other entropy functional with $S(p) = S(1-p)$ $S(0) = 0 = S(1)$ the addition operation in the thermodynamic semiring is a non-associative commutative magma.

For our purposes here we will be choosing an entropy functional that is not the Shannon entropy, so that the addition in the resulting thermodynamic semiring will be commutative but non-associative: the Rényi entropy will be the main example.

We introduce some preliminary notation that will be useful in the following. First, it will be convenient to distinguish between the tree and the bracket notation for a given syntactic object.

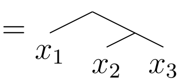
Definition 3.5. *For a syntactic object $T \in \mathfrak{T}_{\mathcal{SO}_0}$, with $\alpha_\ell \in \mathcal{SO}_0$, for $\ell \in L(T)$ the labels at the leaves, we write $\mathcal{B}_T((\alpha_\ell)_{\ell \in L(T)})$ for the "bracket notation" of T. For example:*

(3.3) $T =$  $\longleftrightarrow \quad \mathcal{B}_T(\alpha_1, \alpha_2, \alpha_3) = \{\alpha_1, \{\alpha_2, \alpha_3\}\}$.

Then we want to introduce the relation between full binary trees and elements in a thermodynamic semiring. To do this, we introduce a preliminary step before dealing with the actual syntactic objects, where we consider binary rooted trees with leaves decorated by real numbers (instead of lexical items and syntactic features as in the case of syntactic objects).

We recall here the setting considered in §10 of [39]. Given a full binary rooted tree T with n labelled leaves, where the leaves are labelled by real numbers $x_\ell \in \mathbb{R}$ (or in $\mathbb{R} \cup \{\infty\}$), one can assign to T an n-ary information measure S_T, which is obtained by considering the bracketing $\mathcal{B}_T$ description of the full binary rooted tree as in Definition 3.5. We introduce some notation that will be useful in the following.

Definition 3.6. *We write $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n)$ for the bracketing of the sums $\oplus_{S,\beta}$ of the values x_ℓ at the leaves, associated to the $\alpha_\ell \in \mathcal{SO}_0$, according to the bracketing $\mathcal{B}_T(\alpha_1,\dots,\alpha_n)$ that describes the full binary tree T: for example*

(3.4) $$\mathcal{B}_{T,S,\beta}(x_1,x_2,x_3) := x_1 \oplus_{S,\beta} (x_2 \oplus_{S,\beta} x_3) \quad \textit{for } \; T =$$ 

.

Note that since $\oplus_{S,\beta}$ is commutative, the bracketed sum $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n)$ only depends on T as an abstract tree and does not require the choice of a planar structure.

Given the explicit form (3.2) of the addition $\oplus_{S,\beta}$, we can express the bracketing $\mathcal{B}_T(\alpha_1,\dots,\alpha_n)$ as an optimization over a combination of expressions as in (3.2). To this purpose it is convenient to introduce another notation that keeps track of each of these expressions.

Definition 3.7. *We use the notation $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n;\Lambda)$, with $\Lambda = (\lambda_v)_{v\in V^0(T)}$ a collection of probability distributions $(\lambda_v, 1-\lambda_v)$ at the non-leaf vertices of the tree T, to indicate the expression obtained by recursively computing the expression*

$$\lambda_v x_{v_1} + (1-\lambda_v) x_{v_2} - \beta^{-1} S(\lambda_v)$$

where x_{v_1} and x_{v_2} are the previously computed expression at the the two vertices v_1, v_2 below v.

The following property was proved in §10 of [39]. We rephrase it here according to our present setting and choice of notation.

Proposition 3.8. [39] *Given a binary information measure S that makes the associated thermodynamic semiring $\mathbb{T}_{S,\beta}$ commutative but non-associative, the element $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n) \in \mathbb{T}_{S,\beta}$ associated to a $T \in \mathfrak{T}_{\mathcal{SO}_0}$ as in Definition 3.6 satisfies*

$$\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n) = \min_{\Lambda\in[0,1]^{\#V^o(T)}} \mathcal{B}_{T,S,\beta}(x_1,\dots,x_n;\Lambda)\,.$$

As in (2.3), let $a^T_\ell = P_\ell(\Lambda)$ be the polynomials describing the probability distribution at the leaves of the tree T, determined as in Proposition 2.3 from the $\Lambda = (\lambda_v) \in [0,1]^{\#V^o(T)}$, We write this simply as $A = P(\Lambda)$. There is an n-ary information measure S_T associated to the tree T, which is recursively determined on $T = \mathfrak{M}(T_1,T_2) \in \mathrm{Dom}(h)$ (with a planarization by the head function), with the following property. For a probability distribution $A = (a_1,\dots,a_n)$ at the leaves of T we also define

(3.5) $$\mathcal{B}_{T,S,\beta}(A;x_1,\dots,x_n) := \sum_{\ell\in L(T)} a_\ell\, x_\ell - \beta^{-1} S_T(A)\,.$$

Then, for all $\Lambda \in [0,1]^{\#V^o(T)}$, we have the identity

$$\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n;\Lambda) = \mathcal{B}_{T,S,\beta}(A = P(\Lambda);x_1,\dots,x_n)\,.$$

This also implies the identity

$$\mathcal{B}_{T,S,\beta}(x_1,\ldots,x_n) = \min_{A\in\Delta_{\# L(T)}} \{\sum_\ell a_\ell\, x_\ell - \beta^{-1} S_T(A)\} = \min_{A\in\Delta_{\# L(T)}} \mathcal{B}_{T,S,\beta}(A; x_1,\ldots,x_n)\,.$$

In the example of (3.4) this can be seen explicitly as follows.

Example 3.9. For the case of (3.4), we have two non-leaf nodes $v\in V^o(T)$, hence two probabilities $(\lambda_v, 1-\lambda_v)$. Moreover, we have a probability $A=(a_1,a_2,a_3)$ at the leaves, so we obtain in this case

$$\lambda_2 x_1 + (1-\lambda_2)\lambda_1 x_2 + (1-\lambda_2)(1-\lambda_1)x_3 - \beta^{-1}((1-\lambda_2)S(\lambda_1)+S(\lambda_2))$$
$$= a_1\, x_1 + a_2\, x_2 + a_3\, x_3 - \beta^{-1}(S(\frac{a_2}{1-a_1}) + (1-a_1)S(a_1))\,,$$

with $\lambda_1 = a_2(1-a_1)^{-1}$ and $\lambda_2 = a_1$. Thus, we obtain that the information measure at the leaves, determined by the tree T is

$$S_T(A) = S(\frac{a_2}{1-a_1}) + (1-a_1)S(a_1)\,,$$

and the resulting element that is the image of T in the thermodynamic semiring is given by

$$\mathcal{B}_{T,S,\beta}(x_1,x_2,x_3) = \min_{\lambda_1,\lambda_2\in[0,1]} \{\lambda_2 x_1 + (1-\lambda_2)\lambda_1 x_2 + (1-\lambda_2)(1-\lambda_1)x_3 - \beta^{-1}((1-\lambda_2)S(\lambda_1)+S(\lambda_2))\}$$

$$= \min_{A\in\Delta_2} \{\sum_{i=1}^3 a_i\, x_i - \beta^{-1}(S(\frac{a_2}{1-a_1}) + (1-a_1)S(a_1))\} = \min_{A\in\Delta_2}\{\sum_{i=1}^3 a_i\, x_i - \beta^{-1} S_T(A)\}\,.$$

3.2. **Thermodynamic semirings of functions.** As in [39], in addition to considering the deformations $\mathbb{T}_{S,\beta}$ of the tropical semiring, reviewed above, one can consider semirings of functions, for example continuous functions $\mathcal{C}(X,\mathbb{T})$, with values in the tropical semiring, endowed with the pointwise semiring operations. The thermodynamic deformations of the tropical semiring then induce corresponding deformations $\mathcal{C}(X,\mathbb{T}_{S,\beta})$. We consider here the function space $\mathcal{F}_{S,\beta} := \mathcal{C}(X,\mathbb{T}_{S,\beta})$, endowed with this semiring structure. In particular if the original function space is taken to be $\mathcal{F}=\mathcal{C}(X,\mathbb{R})$, it can be seen as contained inside $\mathcal{C}(X,\mathbb{T})$ and inside $\mathcal{F}_{S,\beta} := \mathcal{C}(X,\mathbb{T}_{S,\beta})$ (with different addition operations and with the original addition playing the role of the semiring multiplication.

We consider here again the given $\varphi : \mathcal{SO}_0 \to \mathcal{F}$ which we now interpret as $\varphi : \mathcal{SO}_0 \to \mathcal{F}_{S,\beta}$, providing a set of functions that encode the lexical items. Instead of considering combinations as elements of $\mathcal{F}\otimes\mathbb{R}[\Lambda]$ as in the previous section, as a way of combining the functions $\varphi(\alpha_\ell)$ according to the tree structure of a syntactic object, we want to use a particular, optimal, choice of λ_v, and correct the combination $\sum_\ell a_\ell \varphi(\alpha_\ell)$ so as to obtain an iterative form of an operation that is non-associative and commutative.

Remark 3.10. One important change, with respect to the construction of §2, is that now the optimality of the choice is achieved *pointwise* in X, resulting in variable $\lambda_v(t)$ that are themselves functions of the point $t\in X$. Thus, the combinations of the wavelet functions $\varphi(\alpha_\ell)$ associated to the lexical items will no longer be linear combinations (i.e. constant coefficients a_ℓ) but combinations of the form $\sum_\ell a_\ell(t)\,\varphi(\alpha_\ell)(t)$, with a probability distribution $A(t)=(a_\ell(t))_{\ell\in L(T)}$ that changes with $t\in X$, and that is determined by a pointwise optimization procedure.

Before describing this procedure in more detail, we recall some property of the information measure that we will use for the thermodynamic semiring.

3.2.1. *The second Rényi entropy.* Some of the results in this section (Theorem 5.1) apply to a broad range of choices of information functional S, for the thermodynamic deformations of the tropical semiring, namely any such functional for which the resulting addition $\oplus_{S,\beta}$ is commutative but non-associative. These include for example, the Renyi entropies Ry_α for any value of $\alpha \neq 1$, or the Tsallis entropies Ts_q for any value $q \neq 1$. (The case at 1 gives back the Shannon entropy for which $\oplus_{S,\beta}$ is both commutative and associative.) However, for the case of Theorem 4.6, we make a specific choice of information measure S, namely a special value of the Renyi entropy for $\alpha = 2$

$$S(p) = \mathrm{Ry}_2(p)\,. \tag{3.6}$$

The second Renyi entropy Ry_2 is also known as the "collision entropy". There are various reasons for making this choice, some conceptual and some practical. The practicality lies in making some computations simpler than in the more general case of arbitrary Ry_α or Ts_q. The more conceptual reason lies in a particularly useful property of the special value Ry_2 of the Rényi entropy, namely the fact that the second Rényi entropy has an interpretation in terms of optimization of redundant non-orthogonal basis expansion (see for instance [29] in signal analysis, and also [7] in quantum information). Moreover, Rényi entropy is computable in brain activity, and has been used in models of visual perception, image segmentation and tracking (for instance [64], [70]). It also has good estimation algorithms (for instance [30], [57], [59]). The connection between entropy (Shannon entropy in this case) and brain signals, in relation to language, was considered in [9], [17].

In the signal analysis setting one has a signal ψ and a finite *dictionary* $\psi_1, \dots \psi_N$ of elementary signals that ψ can be decomposed into. Assuming linearity, this means ψ is (approximately) equal to a linear combination $\sum_i a_i\,\psi_i$. In the nonlinear cases, the combination is replaced by a nonlinear kernel. In general one can assume that the ψ_i are linearly independent, but not necessarily that they are an orthogonal basis with respect to an inner product. The dictionary itself may be very large, but some sparsity assumption is considered. In our case, the dictionary would literally be a "dictionary" consisting of the set $\mathcal{SO}_0$ of lexical items and syntactic features and the set of their images $\varphi(\mathcal{SO}_0)$ inside the space of functions $\mathcal{F}_{S,\beta}$. Since the space $\mathcal{F}_{S,\beta}$ is not endowed with an inner product, we are exactly in the situation where we cannot rely on orthogonality of the basis for signal analysis. In such cases one can introduce a measurement of heterogeneity of the decomposition, a diversity measure that shows the terms of the decomposition are sufficiently dissimilar. An effective such measure is provided by the second Rényi entropy (see [29]).

In our setting, we look at a different perspective, though. We form (pointwise in X) certain combinations of atoms $\varphi(\alpha)$ with $\alpha \in \mathcal{SO}_0$ with coefficients given by a probability distribution $A = (a_\ell)_{\ell\in L}$ (dependent on the point $t \in X$), which depends on a binary rooted tree T with $L = L(T)$ the set of leaves. We can think of the resulting objects $\sum_\ell a_\ell \varphi(\alpha_\ell)$ as a random variable, depending on the probability distribution $A = A(\Lambda)$, with $\Lambda = (\lambda_v)_{v\in V^o(T)}$. Thus, instead of just evaluating the n-ary second Rényi entropy $\mathrm{Ry}_2(A)$ as a diversity measure (as in [29]), we consider the n-ary entropy functionals built from the binary second Rényi entropy and the tree T, namely the $S_T(a_1,\dots,a_n) = \mathrm{Ry}_{2,T}(a_1,\dots,a_n)$, with $n = \#L(T)$. Unlike the n-ary second Rényi entropy, the diversity measure $\mathrm{Ry}_{2,T}$ is adapted to the presence of a fixed tree structure T. Maximization of $\mathrm{Ry}_{2,T}(a_1,\dots,a_n)$ corresponds to minimization of $-\beta^{-1}\mathrm{Ry}_{2,T}(a_1,\dots,a_n)$.

As we discuss below, the choice of an optimal $A = A(\Lambda)$ follows this idea, but instead of just minimizing $-\beta^{-1}\mathrm{Ry}_{2,T}(a_1,\dots,a_n)$, we minimize the combination

$$\mathcal{B}_{T,\mathrm{Ry}_2,\beta}(A, \varphi(\alpha_1),\dots,\varphi(\alpha_n)) = \sum_\ell a_\ell \varphi(\alpha_\ell) - \beta^{-1}\mathrm{Ry}_{2,T}(a_1,\dots,a_n)\,.$$

Note that, since the $\varphi(\alpha_\ell)$ are functions on an underlying space X, the minimization above is pointwise in $t \in X$, hence the resulting $A = A(\Lambda)$ is also a function of $t \in X$, rather than a constant probability distribution, as pointed out in Remark 3.10 above. This means that, this decomposition performed pointwise in $t \in X$ is globally a decomposition with coefficients $a_\ell(t)$ in a ring of functions of $t \in X$. We can then interpret the second Rényi entropy as a measure of diversity for the coefficients $a_\ell(t)$ and the choice of A that is incorporated in the minimization of $\mathcal{B}_{T,\mathrm{Ry}_2,\beta}(A, \varphi(\alpha_1), \dots, \varphi(\alpha_n))$, for fixed $\{\varphi(\alpha_\ell)\}$.

This discussion is only aimed at justifying heuristically our choice of $S = \mathrm{Ry}_2$. The rigorous discussion in the rest of this section is independent of these considerations.

In fact, as mentioned above, for us the main reason for choosing the second Rényi entropy is that the construction described here of a composition of the $\varphi(\alpha_\ell)$ with coefficients given by functions a_ℓ on X, determines an embedding of syntactic objects in the space $\mathcal{F}_{\mathrm{Ry}_2,\beta}$ that is also a morphism of magmas, thus capturing the full algebraic structure of the set of syntactic objects.

We can compute explicitly, for $S = \mathrm{Ry}_2$ the form of the addition $\oplus_{\mathrm{Ry}_2,\beta}$.

Lemma 3.11. *For $S = \mathrm{Ry}_2$, the addition*

$$x \oplus_{S,\beta} y = \min_{\lambda \in [0,1]} \{\lambda x + (1-\lambda) y - \beta^{-1} S(\lambda)\}\,.$$

takes the form

$$x \oplus_{\mathrm{Ry}_2,\beta} y = \min_{\lambda \in [0,1]} \{\lambda x + (1-\lambda) y + \beta^{-1} \log(\lambda^2 + (1-\lambda)^2)\}\,, \tag{3.7}$$

with the minimum achieved at a value $0 < \lambda_{\min}(x,y) < 1$ of the form

$$\lambda_{\min}(x,y) = \frac{u^{-1}}{2}(1 + u - \sqrt{1-u^2}) \qquad \text{for } u = \beta(y-x)/2\,, \tag{3.8}$$

for $x, y \in \mathbb{T}_{S,\beta}$ such that $|y-x| < 2/\beta$. When $|y-x| \geq 2/\beta$ the minimum is at the boundary, $\lambda_{\min}(x,y) \in \{0,1\}$.

Proof. In the case $S = \mathrm{Ry}_2$ we have

$$x \oplus_{\mathrm{Ry}_2,\beta} y = \min_{\lambda \in [0,1]} \{\lambda x + (1-\lambda) y - \beta^{-1} \mathrm{Ry}_2(\lambda)\} = \min_{\lambda \in [0,1]} \{\lambda x + (1-\lambda) y + \beta^{-1} \log(\lambda^2 + (1-\lambda)^2)\}\,.$$

We write

$$F(\lambda) := \lambda x + (1-\lambda) y + \beta^{-1} \log(\lambda^2 + (1-\lambda)^2)\,, \tag{3.9}$$

so that we obtain

$$\frac{\partial}{\partial \lambda} F(\lambda) = x - y + \beta^{-1} \frac{4\lambda - 2}{\lambda^2 + (1-\lambda)^2}$$

and setting $\frac{\partial}{\partial \lambda} F(\lambda) = 0$ we see that, for $y - x \geq 2\beta^{-1}$ we have $\frac{\partial}{\partial \lambda} F(\lambda) < 0$ inside the interval, so the minimum is achieved at the boundary $\lambda = 1$. When $0 < y - x < 2\beta^{-1}$, on the other hand, we have two real solutions, with the smaller one being the minimum. This gives a solution $0 < \lambda_{\min}(x,y) < 1$ of the form

$$\lambda_{\min}(x,y) = \frac{-2\beta^{-1} + x - y + \sqrt{4\beta^{-2} - (x-y)^2}}{2(x-y)} = \frac{u^{-1}}{2}(1 + u - \sqrt{1-u^2}) \tag{3.10}$$

with the change of variable $u = \beta(y-x)/2$, with $0 < u < 1$.

The minimum $\lambda_{\min}(u)$ as in (3.10) satisfies $\lambda_{\min}(u) = 1 - \lambda_{\min}(-u)$ for $|u| \leq 1$. Thus, when $y - x < 0$, we can rewrite the minimization of (3.9) as

$$(1 - \lambda_{\min}(-u))x + \lambda_{\min}(-u) y + \beta^{-1} \log(\lambda_{\min}(-u)^2 + (1 - \lambda_{\min}(-u))^2) \tag{3.11}$$

$$= x + \lambda_{\min}(-u)\, 2\beta^{-1} u + \beta^{-1} \log(\lambda_{\min}(-u)^2 + (1-\lambda_{\min}(-u))^2)\,.$$

We then have, for $y - x \leq -2\beta^{-1}$ the minimum at $\lambda = 0$ and for $-2\beta^{-1} < y - x < 0$ the minimum at $\lambda_{\min}(u)$ as in (3.10). The behavior of the minimum in the range $|u| < 1$ is illustrated in Figure 1. Note the differential singularities at $u = \pm 1$, so that the function is continuous (equal to 0 for $u < -1$ and to 1 for $u > 1$) but only piecewise differentiable. $\square$

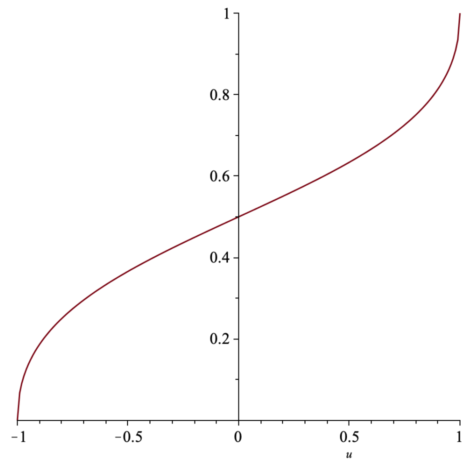


FIGURE 1. The behavior of the function $\lambda_{\min}(x, y)$, in the variable $u = \beta(y - x)/2$ in the range $|u| < 1$.

Remark 3.12. Note that the commutativity property $x \oplus_{\mathrm{Ry}_2,\beta} y = y \oplus_{\mathrm{Ry}_2,\beta} x$ is also ensured by (3.11) since we have $\lambda_{\min}(u)x + (1 - \lambda_{\min}(u))y = (1 - \lambda_{\min}(-u))x + \lambda_{\min}(-u)y$ and we also have $\mathrm{Ry}_2(\lambda) = \mathrm{Ry}_2(1 - \lambda)$.

Definition 3.13. *Given a function $\varphi : \mathcal{SO}_0 \to \mathcal{C}(X, \mathbb{R})$, and assuming $S = \mathrm{Ry}_2$ is the second Rényi entropy, for a given syntactic object $T \in \mathrm{Dom}(h) \subset \mathfrak{T}_{\mathcal{SO}_0}$, with $L = L(T)$ and data $\{\alpha_\ell\}_{\ell \in L}$ at the leaves, the* optimal Λ *is the function*

$$\Lambda = \Lambda_{T, \{\alpha_\ell\}_{\ell \in L(T)}} : X \to [0,1]^{\# V^o(T)}$$

given by the solution of the pointwise optimization

$$\Lambda = \mathrm{argmin}\, \mathcal{B}_{T,S,\beta}(\varphi(\alpha_1), \ldots, \varphi(\alpha_n); \Lambda) = \mathrm{argmin}\, \mathcal{B}_{T,S,\beta}(A = P(\Lambda); \varphi(\alpha_1), \ldots, \varphi(\alpha_n))\,. \tag{3.12}$$

4. Embedding syntactic objects

The construction of Proposition 3.8 answers the question of whether there is a preferred choice of the probabilities $\Lambda \in [0,1]^{\# V^o(T)}$ used to map the tree T to the vector space $\mathcal{F}$ as a combination $\varphi(T)$ of the functions $\varphi(\alpha_\ell)$ at the leaves with coefficients $a^T_\ell = \mathbb{P}_\ell(\Lambda)$.

We now discuss how to use the result of Proposition 3.8 to construct embeddings of syntactic objects inside a thermodynamic semiring of functions as described in §3.2. Namely, we want to construct embeddings of syntactic objects in $\mathcal{C}(X, \mathbb{R})$, in such a way as to ensure that we can also capture the magma operation on syntactic objects, in terms of a commutative, non-associative operation on $\mathcal{C}(X, \mathbb{R})$.

As above let $\mathcal{F}_{S,\beta} = \mathcal{C}(X, \mathbb{T}_{S,\beta})$ be the semiring with the pointwise operations induces by $\mathbb{T}_{S,\beta}$. In general we are interested in the case where $X \subset \mathbb{R}^D$ is a compact region that has a nonempty dense open set $\mathcal{U} \subset X$. In fact, for simplicity, we can just assume (possibly by enlarging the region) that X is a compact D-dimensional smooth manifold without boundary. We then consider the dense subalgebra $\mathcal{C}^\infty(X,\mathbb{R})$ of the algebra $\mathcal{C}(X,\mathbb{R})$ of continuous functions (with the usual pointwise addition and multiplication. We can endow $\mathcal{C}^\infty(X,\mathbb{R})$ with the thermodynamic semiring operations $\oplus_{S,\beta}$, $\odot$ by restriction from viewing $\mathcal{C}^\infty(X,\mathbb{R}) \subset \mathcal{F}_{S,\beta}$. We write $\mathcal{F}^\infty_{S,\beta}$ for $\mathcal{C}^\infty(X,\mathbb{R})$ with these semiring operations.

We view a function $\varphi : \mathcal{SO}_0 \to \mathcal{C}(X,\mathbb{R})$ as above as taking values in $\mathcal{F}_{S,\beta}$. We will also assume here that these functions are in fact smooth, namely $\varphi : \mathcal{SO}_0 \to \mathcal{C}^\infty(X,\mathbb{R})$. By the density of $\mathcal{C}^\infty(X,\mathbb{R})$ in $\mathcal{C}(X,\mathbb{R})$ this can always be achieved by approximation.

Definition 4.1. *Given $\varphi : \mathcal{SO}_0 \to \mathcal{C}^\infty(X,\mathbb{R})$ as above, and $S = \mathrm{Ry}_2$. We define the function $\varphi_{S,\beta} : \mathfrak{T}_{\mathcal{SO}_0} \to \mathcal{F}_{S,\beta}$ in the following way. Given a finite set L and a collection $\{\alpha_\ell\}_{\ell\in L} \subset \mathcal{SO}_0$, set*

$$\varphi_{S,\beta}(T) = \mathcal{B}_{T,S,\beta}(\varphi(\alpha_1),\dots,\varphi(\alpha_n)) = \sum_\ell a_\ell(T)\varphi(\alpha_\ell) - \beta^{-1} S_T(A)\,, \text{for optimal } A = (a_\ell) = P(\Lambda)\,, \tag{4.1}$$

with optimal Λ as in (3.12),

4.1. **Uniform bound and high temperature regime.** We recall here some properties of spaces of functions that we will need in the following. We assume here that the space X is compact. For example, a closed bounded interval $[a,b] \subset \mathbb{R}$ or the product $[a,b] \times \Omega$ of a closed bounded region with nonempty interior, $\Omega \subset \mathbb{R}^D$, typically with $D = 2$ or $D = 3$, representing a region of brain locations and a closed interval of time measurement.

Remark 4.2. For X compact, continuous functions $f \in \mathcal{C}(X,\mathbb{R})$ have a finite maximum and minimum in X. Thus, given a finite set $\{\varphi(\alpha_\ell)\}_{\ell\in L}$ in $\mathcal{F}_{S,\beta} = \mathcal{C}(X,\mathbb{T}_{S,\beta})$ that take values in $\mathbb{R}$ (rather than $\mathbb{R}\cup\{\infty\}$), there is a maximum M_L so that

$$|\varphi(\alpha_\ell)(t)| \le M_L < \infty\,, \tag{4.2}$$

for all $t \in X$ and $\ell \in L$.

Definition 4.3. *For X compact, and for a chosen set L of leaves and corresponding functions $\{\varphi(\alpha_\ell)\}_{\ell\in L}$ in $\mathcal{F}_{S,\beta}$, with M_L the maximum as in (4.2), we choose a $\beta = \beta_L$ with the property that $\beta_L = M_L^{-1}$. The range $\beta < \beta_L$ is the high temperature regime for the syntactic object T with $L(T) = L$ with the given data $\{\varphi(\alpha_\ell)\}_{\ell\in L}$ at the leaves. We can also consider the set $\{\varphi(\alpha)\}_{\alpha\in\mathcal{SO}_0}$, Since $\mathcal{SO}_0$ is a finite set, there is a global maximum $M_{X,\mathcal{SO}_0} < \infty$ for all the $\varphi(\alpha)$ on the compact X, with $M_L \le M_{X,\mathcal{SO}_0}$ for any L, and such that*

$$|\varphi(\alpha)(t)| \le M_{X,\mathcal{SO}_0} < \infty \quad \forall \alpha \in \mathcal{SO}_0\,.$$

We refer to

$$\beta < \beta_{X,\mathcal{SO}_0} = M^{-1}_{X,\mathcal{SO}_0} \tag{4.3}$$

as the global high temperature regime.

Then we obtain the following, as a direct consequence of Definition 4.3.

Lemma 4.4. *In the high temperature regime $\beta < \beta_L$, the estimate $|u(t)| < 1$ holds for all $t \in X$, for*

$$u(t) = \frac{\beta}{2}(\varphi(\alpha_\ell)(t) - \varphi(\alpha_{\ell'})(t)) \tag{4.4}$$

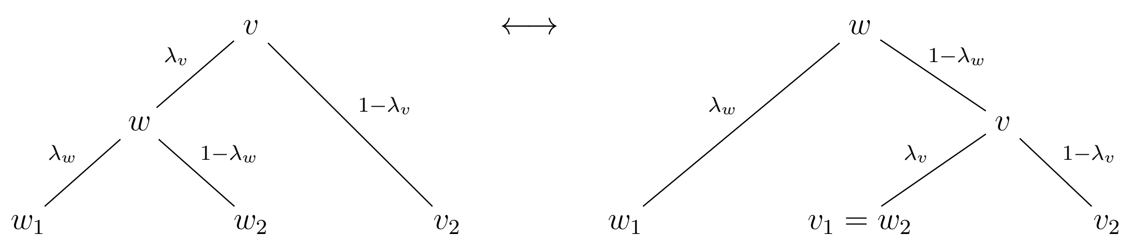
for any $\ell, \ell' \in L$. In the global high temperature regime $\beta < \beta_{X,\mathcal{SO}_0}$ the same $|u(t)| < 1$ estimate holds for
$$u(t) = \frac{\beta}{2}(\varphi(\alpha)(t) - \varphi(\alpha')(t))$$
for any $\alpha, \alpha' \in \mathcal{SO}_0$.

Proof. We have
$$|u(t)| = \frac{\beta}{2}|\varphi(\alpha_\ell)(t) - \varphi(\alpha_{\ell'})(t)| \leq \frac{\beta}{2}(|\varphi(\alpha_\ell)(t)| + |\varphi(\alpha_{\ell'})(t)|) \leq \beta M_{X,\mathcal{SO}_0} < 1\,.$$
$\square$

4.2. **Single wall-crossing.** Any two trees $T \neq T'$ with $L(T) = L(T') = L$ and the same assignment $\{\alpha_\ell\}_{\ell\in L}$ at the leaves, will differ from one another by a series of moves that cross one of the boundary strata in the BHV$_n$ moduli space, for $n = \#L$, see [5]. Namely, if we can assign to both T, T' constant edge length equal to 1 at all edges, moving to a boundary stratum corresponds to shrinking one (or more, for the higher codimension strata of the boundary) of the edges to zero length, hence producing a vertex of higher valence. Crossing a one dimensional boundary stratum corresponds to moving one branching across a vertex, by first shrinking an edge to zero and then creating a new edge of growing length along the other side of the "collision vertex". Any $T \neq T'$ with the same set of labelled leaves differ by repeated application of a number of such operations. We first consider first the case of $T \neq T'$ that differ by a single such operation at one of their non-leaf vertices. We then discuss how to deal with an arbitrary pair $T \neq T'$ with the same data at the leaves.

Theorem 4.5. *Consider a pair of trees $T \neq T'$ with $L(T) = L(T') = L$ and with the same assignment $\{\alpha_\ell\}_{\ell\in L}$, that differ by a single wall crossing in the BHV$_n$ moduli space, with $\#L = n$. Then for such a pair*
$$\varphi_{S,\beta}(T) = \mathcal{B}_{T,S,\beta}(\varphi(\alpha_1), \ldots, \varphi(\alpha_n)) \neq \varphi_{S,\beta}(T') = \mathcal{B}_{T',S,\beta}(\varphi(\alpha_1), \ldots, \varphi(\alpha_n))\,.$$

Proof. In terms of Λ, this transformation of the tree corresponds to the transformation that maps

(4.5)


In the left-hand-side of the diagram (4.5) the coefficients λ_v and λ_w are evaluated at the optimal $\Lambda = \Lambda_{T,\{\alpha_\ell\}}$ and in the right-hand-side at the optimal $\Lambda' = \Lambda_{T',\{\alpha_\ell\}}$. More precisely, since optimization is done one step at a time in the tree, according to the bracketing, with each bracketing computing a single $\oplus_{S,\beta}$ operation, this means that in the left-hand-side of the diagram (4.5) the coefficients λ_v and λ_w realize the optimization that computes the term
$$\begin{aligned}\varphi_{S,\beta}(T_v) &= \big(\mathcal{B}_{T_{w_1},S,\beta}((\alpha_\ell)_{\ell\in L(T_{w_1})}) \oplus_{S,\beta} \mathcal{B}_{T_{w_2},S,\beta}((\alpha_\ell)_{\ell\in L(T_{w_2})})\big) \oplus_{S,\beta} \mathcal{B}_{T_{v_2},S,\beta}((\alpha_\ell)_{\ell\in L(T_{v_2})}) \\ &= (\varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} \varphi_{S,\beta}(T_{w_2})) \oplus_{S,\beta} \varphi_{S,\beta}(T_{v_2})\end{aligned} \tag{4.6}$$
while in the right-hand-side they realize the optimization that computes the term
$$\begin{aligned}\varphi_{S,\beta}(T'_v) &= \mathcal{B}_{T_{w_1},S,\beta}((\alpha_\ell)_{\ell\in L(T_{w_1})}) \oplus_{S,\beta} \big(\mathcal{B}_{T_{v_1},S,\beta}((\alpha_\ell)_{\ell\in L(T_{v_1})}) \oplus_{S,\beta} \mathcal{B}_{T_{v_2},S,\beta}((\alpha_\ell)_{\ell\in L(T_{v_2})})\big) \\ &= \varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} (\varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} \varphi_{S,\beta}(T_{v_2}))\,.\end{aligned} \tag{4.7}$$

Note how the subtrees T_{w_1}, T_{w_2} T_{v_2} are the same in both T and T' hence they map to the same $\varphi_{S,\beta}(T_{w_1})$, $\varphi_{S,\beta}(T_{w_2})$, $\varphi_{S,\beta}(T_{v_2})$ in both (4.6) and (4.7). We then show that

$$(\varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} \varphi_{S,\beta}(T_{w_2})) \oplus_{S,\beta} \varphi_{S,\beta}(T_{v_2}) \neq \varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} (\varphi_{S,\beta}(T_{w_1}) \oplus_{S,\beta} \varphi_{S,\beta}(T_{v_2}))\,, \tag{4.8}$$

for a general choice of the function $\varphi : \mathcal{SO}_0 \to \mathcal{F}_{S,\beta}$. Consider the function $X \to \mathbb{R}^3$ that maps

$$\varphi_v : t \mapsto (x(t), y(t), z(t)) := (\varphi_{S,\beta}(T_{w_1})(t), \varphi_{S,\beta}(T_{w_2})(t), \varphi_{S,\beta}(T_{v_2})(t))\,. \tag{4.9}$$

The condition (4.8) states that there is a nonempty set of $t \in X$ such that the image $\varphi_v(t) = (x(t), y(t), z(t))$ under this map φ_v is not contained in the locus

$$H = \{(x, y, z) \in \mathbb{R}^3 \,|\, x \oplus_{S,\beta} (y \oplus_{S,\beta} z) = (x \oplus_{S,\beta} y) \oplus_{S,\beta} z\}\,. \tag{4.10}$$

The locus (4.10) inside $\mathbb{R}^3$, if non-empty, is of codimension at least one, so it is always possible, up to a small (in the supremum norm in $\mathcal{C}(X, \mathbb{R})$) perturbation, to have a function $\varphi : \mathcal{SO}_0 \to \mathcal{C}(X, \mathbb{R})$ for which the graph of φ_v of (4.9) is not contained in (4.10), hence (4.8) holds. In fact, we can ensure that the set of $t \in X$ for which $\varphi_v(t) \notin H$ is a dense open subset $\mathcal{U}_v$ of X. This gives us that $\varphi_{S,\beta}(T_v) \neq \varphi_{S,\beta}(T'_v)$ at the vertex v where the move described in (4.5) is performed. If T and T' differ only by this single move, the rest of the trees T and T' is then obtained by merging of T_v (respectively, T'_v) with the same $T_{\tilde{v}}$ with $\tilde{v}$ the twin vertex of v in both trees, and then repeatedly merge in other trees (always the same ones for both T and T' until the root is reached). Thus it suffices to show that if $\varphi_{S,\beta}(T_v) \neq \varphi_{S,\beta}(T'_v)$, then also $\varphi_{S,\beta}(T_v) \oplus_{S,\beta} \varphi_{S,\beta}(T_{\tilde{v}}) \neq \varphi_{S,\beta}(T'_v) \oplus_{S,\beta} \varphi_{S,\beta}(T_{\tilde{v}})$. For this one can consider the properties of the successor function of the thermodynamic semiring, as in §9 of [39], namely the function $x \oplus_{S,\beta} 0$. Indeed, we have

$$x \oplus_{S,\beta} y = y + ((x - y) \oplus_{S,\beta} 0) = y + \min_{\lambda}\{\lambda(x - y) - \beta^{-1} S(\lambda)\}\,.$$

We can then see that the successor function $x \mapsto x \oplus_{S,\beta} 0$ is injective as a function $\mathbb{R} \to \mathbb{R}$. For the choice $S = \mathrm{Ry}_2$ of the second Rényi entropy, this can be seen directly from (3.10), using the fact that

$$\frac{\partial}{\partial u} \lambda_{\min}(u) = \frac{1 - \sqrt{1 - u^2}}{2u^2 \sqrt{1 - u^2}} > 0$$

for $|u| < 1$, which holds in the high temperature regime by Lemma 4.4, or from general considerations as in §9 of [39]. The injectivity of the successor function then implies that, for $\varphi_{S,\beta}(T_v) \neq \varphi_{S,\beta}(T'_v)$ we also have

$$(\varphi_{S,\beta}(T_v) - \varphi_{S,\beta}(T_{\tilde{v}})) \oplus_{S,\beta} 0 \neq (\varphi_{S,\beta}(T'_v) - \varphi_{S,\beta}(T_{\tilde{v}})) \oplus_{S,\beta} 0$$

which gives the condition we want, which can then be iterated to obtain $\varphi_{S,\beta}(T) \neq \varphi_{S,\beta}(T')$. We can consider the condition that $\varphi_v(t) \notin H$ for each $v \in V^o(T)$, which ensures that (4.8) holds for the tree T' obtained from T by the same single move at the vertex v. The intersection $\cap_v \mathcal{U}_v$ gives again a dense open set on which all the $\varphi_v(t) \notin H$ conditions hold at once, hence a generic $\varphi : \mathcal{SO}_0 \to \mathcal{C}(X, \mathbb{R})$ for which (4.8) holds for every $v \in V^o(T)$. Thus we obtain $\varphi_{S,\beta}(T) \neq \varphi_{S,\beta}(T')$ whenever T and T' differ by a single move as in (4.5) at any of the vertices. □

We then need to extend this argument to the case where T and T' differ by multiple moves of the type described in (4.5) performed at several vertices. One can iterate the same argument, where at each vertex where a move of type (4.5) happens, one is checking the transversality of a function (obtained from the previous steps) to the same hypersurface (4.10). It is however more convenient to just argue directly in terms of the whole tree using the locus

$$\mathcal{B}_{T,S,\beta}(x_1, \dots, x_n) = \mathcal{B}_{T',S,\beta}(x_1, \dots, x_n)$$

inside $\mathbb{R}^n$, as in Theorem 4.6 below.

4.3. **Transversality and embeddings of syntactic objects.** We then obtain the following general result.

Theorem 4.6. *There is a dense open set of functions $\varphi \in \mathcal{C}^\infty(X,\mathbb{R})^{\mathcal{SO}_0}$, with the property that, for any finite set L with $\#L \leq \#\mathcal{SO}_0$ and any collection $\{\alpha_\ell\}_{\ell\in L} \subset \mathcal{SO}_0$ with $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell' \in L$, the function $\varphi_{S,\beta}$ of Definition 4.1 satisfies $\varphi_{S,\beta}(T) \neq \varphi_{S,\beta}(T')$ for any pair $T \neq T'$ with the same $L(T) = L(T') = L$ and the same data $\{\alpha_\ell\}_{\ell\in L}$ at the leaves.*

Proof. Given the data L and $\{\alpha_\ell\}_{\ell\in L} \subset \mathcal{SO}_0$, we have a function $L \xrightarrow{\alpha} \mathcal{SO}_0 \xrightarrow{\varphi} \mathcal{C}^\infty(X,\mathbb{R})$, hence a continuous map $\alpha^* : \mathcal{C}^\infty(X,\mathbb{R})^{\mathcal{SO}_0} \to \mathcal{C}^\infty(X,\mathbb{R})^L$, equivalently seen as a map $\alpha^* : \mathcal{C}^\infty(X,\mathbb{R}^N) \to \mathcal{C}^\infty(X,\mathbb{R}^n)$, for $N = \#\mathcal{SO}_0$ and $n = \#L$. For $T \neq T'$, with the same $L(T) = L(T') = L$ and $\{\alpha_\ell\}_{\ell\in L}$, The expressions $\varphi_{S,\beta}(T) = \mathcal{B}_{T,S,\beta}(\varphi(\alpha_1),\ldots,\varphi(\alpha_n))$ and $\varphi_{S,\beta}(T') = \mathcal{B}_{T',S,\beta}(\varphi(\alpha_1),\ldots,\varphi(\alpha_n))$ of (4.1) differ by the bracketing of the sequence of binary operations $\oplus_{S,\beta}$ performed at each node of the tree. Since the semiring $\mathbb{T}_{S,\beta}$ is non-associative, *in general*

$$\mathcal{B}_{T,S,\beta}(x_1,\ldots,x_n) \neq \mathcal{B}_{T',S,\beta}(x_1,\ldots,x_n)$$

for a dense open set of $(x_1,\ldots,x_n) \in \mathbb{R}^n$. The locus

$$H_{T,T'} = \{(x_1,\ldots,x_n) \in \mathbb{R}^n \,|\, \mathcal{B}_{T,S,\beta}(x_1,\ldots,x_n) = \mathcal{B}_{T',S,\beta}(x_1,\ldots,x_n)\} \tag{4.11}$$

if non-empty, is of codimension at least one. In fact, for the specific case of $S = \mathrm{Ry}_2$, the second Rényi entropy, we know from the explicit computation (3.10) of the minimizer, that the repeated application of this formula over the sequence of binary operations $\oplus_{S,\beta}$ performed at each node of the tree yields an expression that is almost everywhere a smooth function, except at the vanishing points of the expressions under square roots, where it can have singularities. By the implicit function theorem $H_{T,T'}$ is (away from a lower dimensional singularity locus) a smooth submanifold of $\mathbb{R}^n$ of codimension one (a hypersurface with singularities). By (3.10), the singularities occur when $|u| = 1$, in each application of (3.10) for each $\oplus_{S,\beta}$ involved.

Thus, for each pair $T \neq T'$ we are looking at the relative position of the two maps

$$\begin{tikzcd} X \arrow[r, "\alpha^*(\varphi)"] & \mathbb{R}^n \\ & H_{T,T'} \arrow[u, hook] \end{tikzcd}$$

In general, given a smooth function $\phi : X \to \mathbb{R}^n$, $\phi(x) = (\phi_1(x),\ldots,\phi_n(x))$, with $\phi_i \in \mathcal{C}^\infty(X,\mathbb{R})$, and a subset $H \subset \mathbb{R}^n$ that is a smooth submanifold of positive codimension, the function ϕ is transverse to H if, for every $t \in \phi^{-1}(H)$

$$d_t\phi(T_tX) + T_{\phi(t)}H = T_{\phi(t)}\mathbb{R}^n \simeq \mathbb{R}^n\,. \tag{4.12}$$

Namely, the span of the image of the tangent space of X and the tangent space of H span all of the n dimensions of the ambient space $\mathbb{R}^n$. One writes $\phi \pitchfork H$ for this transversality condition. More generally, for $K \subset X$, one writes $\phi \pitchfork_K H$ to denote the fact that (4.12) holds at all $x \in K \cap f^{-1}(H)$. We also use the notation

$$\pitchfork_K (X,\mathbb{R}^n;H) := \{\phi \in \mathcal{C}^\infty(X,\mathbb{R}^n) \,|\, \phi \pitchfork_K H\}\,. \tag{4.13}$$

The transversality theorem (see [28]) shows that, if H is closed and K is compact, $\pitchfork_K (X,\mathbb{R}^n;H)$ is open and dense in $\mathcal{C}^\infty(X,\mathbb{R}^n)$.

We are here assuming that X is a compact D-dimensional smooth manifold without boundary, where in general $1 \leq D \leq 4$. Thus, except in the case where $n = 3$, we have that D is smaller than both N and n. In the case of $n = 3$, there is a single hypersurface $H_{T,T'}$ corresponding to the condition $(x_1 \oplus_{S,\beta} x_2) \oplus_{S,\beta} x_3 = x_1 \oplus_{S,\beta} (x_2 \oplus_{S,\beta} x_3)$ in $\mathbb{R}^3$. This is the case we have already discussed in Proposition 4.5. Thus, we can assume here that $n \geq 4$ and that $D \leq n$ (hence also $D \leq N$ by hypothesis). Since $D \geq 1$ the transversality theorem can apply, with H a smaooth hypersurface in $\mathbb{R}^n$, since $\dim d_t\phi(T_tX) = D$ and $\dim T_{\phi(t)}H = n-1$ and $D+n-1 \geq n$.

Since the locus $H = H_{T,T'}$ has singularities at which T_xH is not well defined, the transversality condition (4.12) can only hold on the complement of the singular locus of $H_{T,T'}$. A way to avoid this problem is to introduce a small smooth deformation of $H_{T,T'}$ in a small tubular neighborhood in $\mathbb{R}^n$ around the singular locus, and unchanged everywhere else. We denote by $H^\epsilon_{T,T'}$ such a small smooth deformation, depending on a small $\epsilon > 0$. Then $H^\epsilon_{T,T'}$ is a smooth hypersurface in $\mathbb{R}^n$ and the transversality theorem applies, ensuring the existence of an open dense subset of function

$$\mathcal{U}^\epsilon_{T,T'} \subset \mathcal{C}^\infty(X, \mathbb{R}^n)$$

such that $\phi \pitchfork H^\epsilon_{T,T'}$ for all $\phi \in \mathcal{U}^\epsilon_{T,T'}$.

Considering then all the pairs (T,T') with $T \neq T'$ and $L(T) = L(T') = L$ with $\#L = n$, we have

$$\mathcal{U}^\epsilon_n := \bigcap_{T,T'} \mathcal{U}^\epsilon_{T,T'} \subset \mathcal{C}^\infty(X, \mathbb{R}^n)$$

which is a finite intersection of dense open sets, hence itself a dense open set, where transversality simultaneously holds for all the $H^\epsilon_{T,T'}$.

This means that, if we can ensure that $\alpha^*(\varphi) \in \mathcal{U}^\epsilon_n$, by an appropriate choice of $\varphi \in \mathcal{C}^\infty(X, \mathbb{R}^N)$, and for fixed α, we can achieve transversality in all the diagrams

$$\begin{tikzcd} X \arrow[r, "\alpha^*(\varphi)"] & \mathbb{R}^n \\ & H^\epsilon_{T,T'} \arrow[u, hook] \end{tikzcd} \tag{4.14}$$

The map $\alpha^* : \mathcal{C}^\infty(X, \mathbb{R}^N) \to \mathcal{C}^\infty(X, \mathbb{R}^n)$ acts by mapping the elements $\ell \in L$ that label the standard orthonormal basis of $\mathbb{R}^n$ to elements $\alpha_\ell \in \mathcal{SO}_0$, where $\mathcal{SO}_0$ labels the standard orthonormal basis of $\mathbb{R}^N$, and restricting a map $\varphi(t) = (\varphi_\alpha(t))$ to a map $\alpha^*(\varphi)(t) = (\alpha^*(\varphi)_\ell(t)) = (\varphi_{\alpha_\ell}(t))$. Thus, the preimage

$$(\alpha^*)^{-1}(\mathcal{U}^\epsilon_n) = \{\varphi = (\varphi_\alpha) \,|\, (\varphi_{\alpha_\ell})_{\ell\in L} \in \mathcal{U}^\epsilon_n\} \subset \mathcal{C}^\infty(X, \mathbb{R}^N)$$

is the subset on which transversality in all the diagrams (4.14) is achieved. If the data $\{\alpha_\ell\}_{\ell\in L}$ satisfy $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell' \in L$, then the map $\alpha^* : \mathcal{C}^\infty(X, \mathbb{R}^N) \to \mathcal{C}^\infty(X, \mathbb{R}^n)$ is just a projection onto the α_ℓ coordinates of $\varphi = (\varphi_\alpha)$, hence it is an open map, so the preimage of a dense set is dense. Thus, for all $n \leq N$, with any choice of data $\{\alpha_\ell\}_{\ell\in L}$ satisfy $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell' \in L$, we have $(\alpha^*)^{-1}(\mathcal{U}^\epsilon_n)$ a dense open set in $\mathcal{C}^\infty(X, \mathbb{R}^N)$, and the intersection of these (finitely many) dense open sets is still a dense open set on which all these transversality conditions are simultaneously realized. □

This argument is limited to the case where $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell' \in L$, which in particular implies $n \leq N$, imposing a finite cap on the size of the syntactic objects $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with $\#L(T) \leq N$.

If we allow cases where $\alpha_\ell = \alpha_{\ell'}$ for some $\ell \neq \ell' \in L$, then the map $\alpha^* : \mathcal{C}^\infty(X, \mathbb{R}^N) \to \mathcal{C}^\infty(X, \mathbb{R}^n)$ is a projection followed by an embedding of the image of this projection as a diagonal in $\mathcal{C}^\infty(X, \mathbb{R}^n)$ and embeddings are open maps only when their image is open (which is not the case here), so the preimage $(\alpha^*)^{-1}(\mathcal{U}^\epsilon_n)$ will no longer be a dense set. In fact, it is clear that, if some conditions $\alpha_\ell = \alpha_{\ell'}$ hold, there can be some trees $T \neq T'$ that may no longer be distinguishable.

So this presents two problems from the point of view of encoding syntactic objects: one is the cap on the size by $\#L(T) \leq N$ and one is (even for trees that meet this size constraint) the fact that we cannot accommodate the case of some $\alpha_\ell = \alpha_{\ell'}$.

These two issues can be analyzed by identifying the kind of syntactic objects where these problems would arise.

4.4. **Repetitions and copies.** The assumption that $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell'$ in L, used in Theorem 4.6, is not realistic from the linguistic viewpoint, as one certainly has syntactic objects where some of the lexical items are *repetitions*, as in "*John meets John*", where the two Johns are understood to not be the same. However, repetitions are not *copies*, hence one can distinguish them as in $John_1$ and $John_2$ and assume that they are stored as different $\varphi(\mathrm{John}_1) \neq \varphi(\mathrm{John}_2)$. So repetitions can be accommodated within the setting of Theorem 4.6.

A different issue arises in the case of *copies*, namely elements $\alpha_\ell, \alpha_{\ell'}$ that are actually the same. In the linguistics literature one expresses this in terms of a FormCopy function that has the effect we described above, to restrict two different coordinates $\alpha_\ell, \alpha_{\ell'}$ with $\ell \neq \ell'$ to lie on the diagonal where $\alpha_\ell = \alpha_{\ell'}$. This occurs in important situations in syntax, such as Obligatory Control, Parasitic Gap, Across-the-Board movement. With the mathematical model of [35], it does *not* occur in movement by Internal Merge, because in the coproduct terms of the form $T_v \otimes T/T_v$ the extraction of the accessible term T_v is accompanied by the deletion of the deeper copy in T/T_v so no FormCopy is required. However, it does occur in the other mentioned cases, for example for Obligatory Control, as discussed in §3.8.2 of [35].

In the mathematical formulation of these cases involving FormCopy, one performs an identification of the copies, and the coproduct then acts as the resulting coproduct of a Hopf algebra of graphs, rather than as the coproduct on trees by extracting and quotienting the identified terms as a single subgraph. In particular, this identification means that, in such cases, we are embedding graphs obtained from trees by performing these identifications.

In the setting considered here, the problem are those functions

$$\varphi : X \to \Delta_{\ell,\ell'} \subset \mathbb{R}^n$$

where $\Delta_{\ell,\ell'} = \{(x_1, \ldots, x_n \in X^n \,|\, x_\ell = x_{\ell'}\}$ is the diagonal, so that $\varphi_\ell(t) = \varphi_{\ell'}(t)$ for all $t \in X$, or more generally functions

$$\varphi : X \to \Delta_{\ell_1,\ldots,\ell_k} \subset \mathbb{R}^n$$

for one of the deeper diagonals

$$\Delta_{\ell_1,\ldots,\ell_k} = \{(x_1, \ldots, x_n \in X^n \,|\, x_{i_1} = \cdots = x_{i_k}\}\,.$$

We can think of $\mathbb{R}^n$ as parameterizing the position of an ordered set of n points in $\mathbb{R}$, and the diagonals as the loci where a certain number of these points collide together.

A way to handle these cases within the formalism considered here, would be to replace these functions with function that contain more information about such colliding points. This can be done, for example by resolving two coincident points as two colliding points that are infinitesimally displaced along a certain direction. This extra datum can be seen as a choice of a way to push the colliding points apart.

This idea can be formalized geometrically. There is a mathematical construction that is precisely designed to the purpose of resolving the diagonals in this way in configurations of n points in an ambient space Y. This is known as the Fulton-MacPherson compactification of the configuration space and it is constructed algebro geometrically, [26]. Given a smooth projective algebraic variety Y (in our case here it would be just the projective line $Y = \mathbb{P}^1$) one considers the complement of the diagonals

$$\mathrm{Conf}_n(Y) = Y^n \smallsetminus \bigcup_{\ell,\ell'} \Delta_{\ell,\ell'}$$

and a compactification

$$Y[n] = \overline{\mathrm{Conf}_n(Y)}$$

of this space, obtained as a sequence of blowups of diagonals in Y^n. There is a projection map $\pi_n : Y[n] \twoheadrightarrow Y$. Under this projection, the preimage $\pi_n^{-1}\mathrm{Conf}_n(Y) \cong \mathrm{Conf}_n(Y)$ so we can view the configuration space as sitting inside $Y[n]$ and the complement $Y[n] \smallsetminus \mathrm{Conf}_n(Y)$ is a normal crossings divisor (a union of hypersurfaces intersecting transversely). These hypersurfaces and their intersections are the boundary strata of the compactification $Y[n]$ of $\mathrm{Conf}_n(Y)$. The details of this construction can be found in [26] and are not needed here. A more differential geometric formulation of these compactifications is described in [8]. In our setting we consider this space for the real projective line $Y = \mathbb{P}^1(\mathbb{R})$.

Intuitively, the effect of this compactification is to replace all the diagonals where two or more of the coordinates agree, with a projective bundle where the projective spaces that replace each point of the diagonal represent the different "directions" with which the colliding points approach the diagonal. For some considerations about working with real rather than complex spaces see for instance the discussion in [71]. These compactifications also have a description in terms of operads, see [40].

Thus, given maps as above, with

$$\varphi : X \to \Delta_{\ell,\ell'} \hookrightarrow \mathbb{R}^n \hookrightarrow \mathbb{P}^1(\mathbb{R})^n$$

or similarly for more general diagonals, we can choose a local section $\sigma : \mathbb{P}^1(\mathbb{R})^n \to \mathbb{P}^1(\mathbb{R})[n]$ with $\pi_n \circ \sigma = \mathrm{id}$, over the domain $\mathrm{Dom}(\sigma) = \mathbb{R}^n \hookrightarrow \mathbb{P}^1(\mathbb{R})^n$. This choice provides us with a lift

$$\begin{tikzcd} & & \mathbb{P}^1(\mathbb{R})[n] \arrow[d, "\pi_n"] \\ X \arrow[urr, "\sigma\circ\varphi"] \arrow[r, "\varphi"'] & \mathbb{R}^n \arrow[r, hook] & \mathbb{P}^1(\mathbb{R})^n \end{tikzcd} \tag{4.15}$$

where the previously colliding $\varphi_\ell = \varphi_{\ell'}$ now become separated along the direction specified by the section σ.

Thus, in this model we can think of any FormCopy operation that is perfomed on a syntactic object, which causes some identifications $\alpha_\ell = \alpha_{\ell'}$ as consisting not just of the restriction to diagonals $\Delta_{\ell,\ell'}$ (as formulated in §3.8.2 of [35]) but as incorporating the additional datum of the section σ, which lifts $\Delta_{\ell,\ell'}$ to $\sigma(\Delta_{\ell,\ell'}) \subset \mathbb{P}^1(\mathbb{R})[n]$, so that it results in a restriction to the boundary divisors of the Fulton-MacPherson compactification of $\mathrm{Conf}_n(\mathbb{R})$, where points do not collide but maintain mutual ϵ-dispacements (with $\epsilon \to 0$) in a direction specified by σ.

4.5. **Size constraints and performance.** We can then also address the problem of the bounded size $\#L \leq \#\mathcal{SO}_0$ assumed in Theorem 4.6. One of the main properties of the computational structure of syntax is that it generates a countably infinite set of possible structures (syntactic objects) our of a finite lexicon. While there are performance constraints in our brains, we immediately understand that certain recursions in syntax exist and would produce syntactically valid sentences of arbitrarily large size.

The size cap $\#L \leq \#\mathcal{SO}_0$ was imposed by the necessity to use only assignments of lexical items and features at the leaves with $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell'$ in L. That problem can be bypassed, as we've been arguing above by resolving the diagonals to the boundary divisors of the Fulton-MacPherson compactification.

However, there is a way in which one does see the performance problem arise in this representation. It is in the use of uniform bounds and the high-temperature regime discussed in §4.1.

As we have seen in §4.1, for X a compact manifold, one can find a uniform bound for the norm

$$\sup_{\alpha \in \mathcal{SO}_0} \sup_{t \in X} |\varphi(\alpha)(t)| \leq M_{X,\mathcal{SO}_0} < \infty.$$

The existence of this bound ensures that there is a high temperature range in which the function $u(t)$ of (4.4) satisfies $\sup_{t\in X} |u(t)| < 1$, so that the optimization value $\lambda_{\min}(\varphi(\alpha)(t), \varphi(\alpha')(t))$ used in computing $\varphi(\alpha)(t) \oplus_{S,\beta} \varphi(\alpha')(t)$ of (3.10) is contained in $(0, 1)$. The fact that the minimum occurs in the interior of the interval guarantees that the resulting $\varphi(\alpha)(t)\oplus_{S,\beta}\varphi(\alpha')(t)$ still contains the information of both $\varphi(\alpha)(t)$ and $varphi(\alpha')(t)$.

When we consider iterated applications of $\oplus_{S,\beta}$ rather than a single one, in the computation of $\varphi_{S,\beta}(T)$, at each step the optimal $\lambda_{\min}(x, y)$ of (3.10) is computed for more general $x = \varphi_{S,\beta}(T_1)$ and $y = \varphi_{S,\beta}(T_1)$ for subtrees T_1, T_2 of T. Thus, the condition that

$$\lambda_{\min}(\varphi_{S,\beta}(T_1)(t), \varphi_{S,\beta}(T_1)(t)) \in (0, 1)$$

now depends on the condition that

$$u(t) = \frac{\beta}{2}(\varphi_{S,\beta}(T_1)(t) - \varphi_{S,\beta}(T_1)(t))$$

satisfies $\sup_{t\in X} |u(t)| < 1$. This no longer follows directly from the fact that the functions $\varphi(\alpha)$ satisfy the uniform bound. For each size $n = \#L$, one does have a uniform bound

$$\sup_{t\in X} |\varphi_{S,\beta}(T)(t)| \leq M_{X,\mathcal{SO},n} < \infty$$

for all $T \in \mathfrak{T}_{\mathcal{SO}_0}$ with $\#L(T) \leq n$. However, for $n \to \infty$ one cannot guarantee a uniform bound, and typically one expects that $M_{X,\mathcal{SO},n} \to \infty$ as $n \to \infty$. This in turn implies that, as larger and larger sizes of syntactic objects are considered, the high-temperature regime

$$\beta < \beta_{X,\mathcal{SO},n} := M^{-1}_{X,\mathcal{SO},n}$$

with shrink as $\beta_{X,\mathcal{SO},n} \to 0$ for $n \to \infty$. This need to progressively shrink β (increasing the termperature parameter β^{-1}) signals what can be regarded as accounting, in this model, for a loss of performance in the handling increasingly large syntactic objects, in addition to the accumulating computational costs of performing the iterated operations $\oplus_{S,\beta}$.

5. The magma operation

Next we can show that, not only the mapping of syntactic objects to a wavelet encoding in the space $\mathcal{F}_{S,\beta}$ is faithful (at least using for S the second Rényi entropy) but it also is compatible with the magma operation (for more general choices of S). Namely, the magma operation on syntactic objects maps to the semiring addition in $\mathcal{F}_{S,\beta}$.

Theorem 5.1. *Consider a choice of information function S such that the addition $\oplus_{S,\beta}$ of the thermodynamic semiring $\mathbb{T}_{S,\beta}$ is commutative but non-associative (for example the Rényi entropy considered above), hence so is $\mathcal{F}_{S,\beta}$. The map $\varphi_{S,\beta} : \mathfrak{T}_{\mathcal{SO}_0} \to \mathcal{F}_{S,\beta}$ as in* (4.1)*, for Λ satisfying the optimal* (3.12)*, extended as discussed in §4.4, is a morphism of commutative non-associative magmas.*

Proof. The deformed addition $\oplus_{S,\beta}$ gives $\mathcal{F}_{S,\beta}$ the structure of commutative non-associative magma, which is unital with unit 0. The unit 1 (the formal empty tree) of the magma $\mathfrak{T}_{\mathcal{SO}_0}$ maps to 0 under $\varphi_{S,\beta}$. To see that the map is compatible with the magma operation, namely

$$\varphi_{S,\beta}(\mathfrak{M}(T_1, T_2)) = \varphi_{S,\beta}(T_1) \oplus_{S,\beta} \varphi_{S,\beta}(T_2)\,,$$

we need to compare

$$\min_{\lambda_v,\Lambda_1,\Lambda_2}\{ \sum_{\ell\in L(T_1)} \lambda_v a_\ell(\Lambda_1) + \sum_{\ell'\in L(T_2)} (1 - \lambda_v)a_{\ell'}(\Lambda_2) - \beta^{-1}S_{\mathfrak{M}(T_1,T_2)}(\lambda_v a_\ell(\Lambda_1), (1 - \lambda_v)a_{\ell'}(\Lambda_2))\}$$

with

$$\min_{\lambda_v}\{\lambda_v \min_{\Lambda_1}\{\sum_{\ell\in L(T_1)} a_\ell(\Lambda_1) - \beta^{-1} S_{T_1}(A(\Lambda_1))\}$$

$$+(1-\lambda_v)\min_{\Lambda_2}\{\sum_{\ell'\in L(T_2)} a_{\ell'}(\Lambda_2) - \beta^{-1} S_{T_2}(A(\Lambda_2))\} - \beta^{-1} S(\lambda_v, 1-\lambda_v)\}\,.$$

The recursive structure of the entropy functionals S_T shows that (Lemma 10.3 of [39])

$$S_{\mathfrak{M}(T_1,T_2)}(\lambda_v a_\ell(\Lambda_1), (1-\lambda_v) a_{\ell'}(\Lambda_2)) = S(\lambda_v, 1-\lambda_v) + \lambda_v S_{T_1}(A(\Lambda_1)) + (1-\lambda_v) S_{T_2}(A(\Lambda_2))\,.$$

The identification of the two expressions above then follows, by observing that one can separate the minimization with respect to Λ_1, Λ_2 and λ_v. □

6. Encoding Merge

The magma operation $\mathfrak{M}(T, T')$ on syntactic objects suffices to account for structure building by External Merge, but movement requires also an Internal Merge operation, which can be formulated as in [35] in terms of a Hopf algebra of workspaces where the coproduct extracts accessible terms T_v inside a syntactic object T. The Internal Merge then forms the new structure $\mathfrak{M}(T_v, T/T_v)$.

By Theorem 5.1 we know that

$$\varphi_{S,\beta}(\mathfrak{M}(T_v, T/T_v)) = \varphi_{S,\beta}(T_v) \oplus_{S,\beta} \varphi_{S,\beta}(T/T_v)\,,$$

where the two terms are given by

$$\varphi_{S,\beta}(T_v) = \mathcal{B}_{T_v,S,\beta}((x_\ell)_{\ell\in L(T_v)}) \quad \text{and} \quad \varphi_{S,\beta}(T/T_v) = \mathcal{B}_{T/T_v,S,\beta}((x_\ell)_{\ell\in L(T/T_v)}).$$

Note that, however, this does not provide us with a procedure for computing $\varphi_{S,\beta}(T_v)$ from $\varphi_{S,\beta}(T)$ since, the addition $\oplus_{S,\beta}$ not being invertible in the semiring, one cannot directly extract $\varphi_{S,\beta}(T_v)$ from $\varphi_{S,\beta}(T)$ simply via the semiring operations. The extraction of accessible terms and their representations has to come from additional algebraic structure (as we already know, from the use of Hopf algebras to model this extraction procedure at the level of the syntactic objects and workspaces in [35]).

6.1. **Algebra over an operad.** Thus, we first describe what additional algebraic structures we have available on $\mathcal{F}_{S,\beta}$ and $\varphi_{S,\beta}(\mathfrak{T}_{\mathcal{SO}_0}) \subset \mathcal{F}_{S,\beta}$. A first thing we can observe is that, just like the set of syntactic objects $\mathfrak{T}_{\mathcal{SO}_0}$ is an algebra over the Merge operad (see [35]), its image $\varphi_{S,\beta}(\mathfrak{T}_{\mathcal{SO}_0})$ also has the same structure. These algebra over an operad structure of syntactic objects is useful for important purposes, including modeling the assignment of theta roles for semantics, [36], and the structure of head, complement and phases. So it is important that the same structure is maintained in the representation of syntactic objects in $\mathcal{F}_{S,\beta}$.

Lemma 6.1. *Let $\mathcal{M}$ denote the Merge operad of non-planar binary rooted trees (with unlabeled leaves), with the operad compositions $\gamma(T;(T_\ell)_{\ell\in L(T)})$ that graft the root vertex of each T_ℓ into the leaf $\ell \in L(T)$ of T. The image $\varphi_{S,\beta}(\mathfrak{T}_{\mathcal{SO}_0}) \subset \mathcal{F}_{S,\beta}$ is an algebra over the operad $\mathcal{M}$.*

Proof. We consider $T \in \mathcal{M}(n)$ as an operation with $n = \#L(T)$ inputs and one output. We write this operation in the form $\mathcal{B}_{T,S,\beta}(x_1,\ldots,x_n)$, where we consider $x_1,\ldots,x_n$ as input variables in $\mathcal{F}_{S,\beta}$. For example, $x \oplus_{S,\beta} y$ is the operation (cherry tree) in $\mathcal{M}(2)$ that takes inputs $x, y \in \mathcal{F}_{S,\beta}$ and provides an output that is also in $\mathcal{F}_{S,\beta}$. This gives $\mathcal{F}_{S,\beta}$ the structure of an algebra over the Merge operad $\mathcal{M}$, with $\varphi_{S,\beta}(\mathfrak{T}_{\mathcal{SO}_0}) \subset \mathcal{F}_{S,\beta}$ a subalgebra over the same operad, since for $x_i = \varphi_{S,\beta}(T_i)$ the result $\gamma(T;(\varphi_{S,\beta}(T_\ell))_{\ell\in L(T)})$ is also in $\varphi_{S,\beta}(\mathfrak{T}_{\mathcal{SO}_0})$. □

Note that, here, we consider $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n)$ as an operation with input varables x_i. It is only when these variables are specialized to chosen inputs $x_i = \varphi_{S,\beta}(T_i)$ with $T_i \in \mathfrak{T}_{\mathcal{SO}_0}$ (including the case of $x_i = \varphi_{S,\beta}(\alpha_i)$ with $\alpha_i \in \mathcal{SO}_0$), that $\mathcal{B}_{T,S,\beta}(\varphi_{S,\beta}(T_1),\dots,\varphi_{S,\beta}(T_n))$ is actually evaluated by computing the optimizing $A(\Lambda)$ as in (3.12).

Here one should interpret the operations $\mathcal{B}_{T,S,\beta}(x_1,\dots,x_n)$ as *circuits* that input certain waves $x_i = \varphi_{S,\beta}(T_i)$ (functions in $\mathcal{C}(X,\mathbb{R})$) and output a now "combined" wave form. We know, as discussed above, that these circuits are all obtained by compositions (according to the structure of the tree T) of one simple *gate* that maps a pair of inputs x, y to the output $x \oplus_{S,\beta} y$. As we discuss in §6.2 below, we can then think of the Merge action as an action that *transforms these circuits* into new circuits.

6.2. **Workspaces, coproduct, and Merge.** We then discuss how to implement the Hopf algebra structure that gives the extraction of accessible terms, and the Merge action as a Hopf algebra Markov chain.

Definition 6.2. *Consider first the set $\mathfrak{B}_{S,\beta}$ of all the operations $\mathcal{B}_{T,S,\beta} : \mathcal{F}_{S,\beta}^{\#L(T)} \to \mathcal{F}_{S,\beta}$. We form the vector space $\mathcal{V}_{S,\beta} = \mathrm{span}_{\mathbb{R}}(\mathfrak{B}_{S,\beta})$ spanned by this set. We then form the symmetric algebra over this vector space, which we can identify with the polynomial algebra in the $\mathfrak{B}_{S,\beta}$,*

$$\mathcal{W}_{S\beta} := \mathrm{Sym}(\mathcal{V}_{S,\beta}) = \mathbb{R}[\mathfrak{B}_{S,\beta}]\,.$$

We refer to the set $\mathfrak{B}_{S,\beta}$ as the set of thermodynamic syntax circuits *with thermodynamic datum (S,β), and to $\mathcal{W}_{S\beta}$ as the* space of thermodynamic syntax circuits*.*

Theorem 6.3. *The space $\mathcal{W}_{S\beta}$ of thermodynamic syntax circuits has a bialgebra (Hopf algebra) structure that is isomorphic to the one defined on workspaces. The Merge action on workspaces is faithfully realized by a Hopf algebra Markov chain in $\mathcal{W}_{S\beta}$.*

Proof. For a forest $F = T_1 \sqcup \cdots \sqcup T_N$, we write

$$\mathcal{B}_{F,S,\beta} := \mathcal{B}_{T_1,S,\beta} \cdots \mathcal{B}_{T_N,S,\beta}$$

by interpreting the monomial as a basis element in $\mathcal{W}_{S\,\beta}$. The elements $\mathcal{B}_{F,S,\beta}$ are the representations of workspaces F.

It is then possible to endow $\mathcal{W}_{S\beta}$ with a coproduct and a bialgebra structure (or Hopf algebra, depending on slightly different forms of the coproduct, as discussed in §1.s of [35]) using the same coproduct defined on workspaces in [35], by setting

$$\Delta(\mathcal{B}_{F,S,\beta}) = \mathcal{B}_{F,S,\beta} \otimes 1 + 1 \otimes \mathcal{B}_{F,S,\beta} + \sum_{F_{\underline{v}}} \mathcal{B}_{F_{\underline{v}},S,\beta} \otimes \mathcal{B}_{F/F_{\underline{v}},S,\beta}\,. \tag{6.1}$$

We then represent the Merge action in the form

$$\mathfrak{M}^{(S,\beta)}_{T,T'} = \mu \circ ((- \oplus_{S,\beta} -) \otimes \mathrm{id}) \circ \delta_{T,T'} \circ \Delta\,, \tag{6.2}$$

where μ is the multiplication in $\mathbb{R}[\mathfrak{B}_{S,\beta}]$ and $\delta_{T,T'}$ is the identity on terms of the coproduct of the form $\mathcal{B}_{T,S,\beta}\mathcal{B}_{T',S,\beta} \otimes \mathcal{B}_{F',S,\beta}$ and zero on all other terms. The notation $(- \oplus_{S,\beta} -)$ stands for the binary operation $\oplus_{S,\beta}$ that takes input $\mathcal{B}_{T_1,S,\beta}\mathcal{B}_{T_2,S,\beta}$ and computed $\mathcal{B}_{T_1,S,\beta}(x_1,\dots,x_n) \oplus_{S,\beta} \mathcal{B}_{T_2,S,\beta}(y_1,\dots,y_m)$. These Merge operations can be assembled into a Hopf algebra Markov chain (as discussed in §1.9 of [35]) of the form

$$\mathcal{K} = \mu \circ ((- \oplus_{S,\beta} -) \otimes \mathrm{id}) \circ \Pi_{(2)} \circ \Delta\,, \tag{6.3}$$

where $\Pi_{(2)}$ is the projection of $\mathcal{W}_{S,\beta} \otimes \mathcal{W}_{S,\beta}$ onto the subspace spanned by elements of the from $\mathcal{B}_{T_1,S,\beta}\mathcal{B}_{T_2,S,\beta} \otimes \mathcal{B}_{F',S,\beta}$ for some arbitrary forest F' and arbitrary trees T_1, T_2. ☐

7. Realization as phase synchronizations

Our goal in this section is to make the previous discussion slightly more concrete, by showing that a form of the embodiment of Merge on a non-associative commutative thermodynamic semiring addition can be implemented on an especially simple and explicit class of wavelets, in the form of phase synchronizations on superpositions of sinusoidal waves.

This very simple realization is again in many ways an abstract toy model, but it has the advantage of dealing with objects (sinusoidal waves) and operations (phase locking, cross-frequency synchronization, phase-amplitude coupling) that are widely used in the neurocomputational context. Oscillations are a way of describing neural activities, and modulations of oscillatory activity can be measured for perceptual and cognitive functions, linguistic and otherwise.

A large amount of work in language neuroscience has pointed to phase synchronization, phase coupling, phase-amplitude coupling (PAC) being modulated in speech and language processing, [2], [3], [10], [15], [16], [19], [34], [41] [48], [50], [51], [58], [60], [62], [63], [66], [68], [74], [75]. It is natural to expect to see a signature of the hierarchical structure formation of syntax reflected in such operations on wavelets, though a lot of the literature focuses on PAC effects for speech listening. Empirical results linking PAC and phase synchronization to syntax include [1], [10], [73], while empirical results on the relation between synchronization and syntax are discussed in [49]. It is in general a very difficult problem to directly relate observable wavelet data from oscillations to an underlying algorithmic computational models embodied and implemented in neural population activity. In this sense the realization we have described of Merge as a semiring addition on a space of wavelets is limited, as it does not address how these operations would be encoded in receptive fields and population activity.

Our mathematical model here is not yet designed, at this stage, to fit any particular theoretical model of language neuroscience, or proposal for the interpretation of the relation between syntax and brain waves, with synchronization and modulation operations.

We will, however, recall briefly some more specific examples of recent work that shares some conceptual aspects with the model we will describe in this section, based on an operation of synchronization.

7.1. **Martin's compositional neural architecture proposal.**

In [43], [44], [45], a computational model of low frequency brain oscillations generating linguistic structure was developed, and a related theoretical framework was developed in [42]. The model is centered on time-based binding operations, which means that in a neural ensemble the relative time of firing can encode the relation between different patterns in the system, with temporal proximity and phase synchronization reflect relations of word level inputs into higher hierarchical structures. This process of structure building or parsing can be modeled by a network where nodes at a higher level show more phase synchronization encoding structural relations in phrases and sentences. This "rhythmic computation" model uses time and synchronization/desynchronization to carry information about hierarchical structures in language. In this model a phrase or sentence can be formed, starting from a vector representation of the input words via a conjoining operation resulting in a conjunctive code on a higher level of the network. In particular, it is shown in [1] that the level of synchronization and power connectivity is higher for sentences than for phrases, for comparable semantic and statistical properties. The precise nature of these synchronization operations and whether they satisfy the same algebraic properties (commutativity and non-associativity) of the syntactic Merge are not explicitly discussed in [42], but the kind of circuits that we discussed in §6.1 above are in principle compatible with this kind of model. Other aspects of this model include the interplay between slower rhythm oscillations and higher frequencies providing a modulation as part of the generation of higher-level linguistic structures.

An interesting observation made in [42] on the nature of the structure building operations is that words and phrased compose additively and not through a multiplicative operation (like a tensor product). It is posed as a question there whether vector addition (linear or convex combinations) would be sufficient. What we have shown in this paper confirms that the Merge operation on vector representations (in a vector space of functions) of lexical items is indeed not a tensor product type operation but an addition-type operation. However, we have also shown here that simple vector addition, as in convex combinations, is not sufficient, as we discussed in §2.3. On the other hand, a modification of a convex combination of vectors, performed via an optimization of such combinations involving an entropy functional, provides an additive operation (the addition in a thermodynamic semiring) that has all the right algebraic properties required for compositionality in natural language. In this section we will also show that such an operation is implementable in the form of a synchronization.

7.2. **Murphy's ROSE proposal.** In [54], [56] Murphy proposes a possible theoretical model, called ROSE, for neurocomputational realizations of Merge, based on neurosurgery experiments measuring phase shifts towards increased phase synchronization and modulation effects of brain waves. In essence, Murphy's ROSE model (Representation, Operation, Structure, Encoding) proposes a multi-step process, with a first process of spike-phase coupling that bundles atomic features together (R and O part), followed by phase-coupling of a slower theta rhythm with this morphological spike-phase complex (implementing lexical access and memory) where spikes (encoding features) couple with gamma phases (encoding morphemes) and the amplitude of this gamma wave couples to a theta phase (encoding lexical items). In this part of the process (S and E part) lexical items are encoded as theta-gamma-spike synchronized complexes, spatially separated as different cellular clusters that oscillate in the theta range Structure formation in syntax is then modeled as coupling to delta waves, which involves a delta phase-phase locking for merging, while delta phase-locking with the amplitude both of these theta rhythms implement the presence of a head function for syntactic objects.

For the purposes of the present paper, we do not analyze these models in any detail, as we are not trying to formalize or validate any specific existing model. We just point out one particular aspect, that is more directly relevant to the discussion of the previous sections, namely the modeling of hierarchical structure building based on phase synchronization and cross-frequency coupling, which is what motivates and inspires the construction described in this section.

Indeed, our goal here is to show that, in our mathematical model, if lexical items are encoded through simple sinusoidal waves, then a model of Merge as discussed in the previous sections can be implemented by synchronizing phases via cross-frequency synchronization, where the phases are locked to a common weighted average expressed as a sum $\oplus_{\mathrm{Ry}_2,\beta}$. The operation we describe is different from what is proposed in [54], [56], where it is suggested that a mechanism for bracketing would rely on time-ordering, but a direct algebraic verification of nonassociativity is not directly available. In the construction we discuss here nonassociativity is realized implementing a phase synchronization via semiring addition.

7.3. **The successor function in thermodynamic semirings.** As shown in §9 of [39], the algebraic properties of the thermodynamic semiring $\mathbb{T}_{S,\beta}$ can be encoded in its *successor function* (the Legendre transform of the entropy functional $\beta^{-1}S$). It is defined as

$$\Upsilon(x,\beta) = x \oplus_{S,\beta} 0 = \min_{\lambda\in[0,1]} \{\lambda x - \beta^{-1} S(\lambda)\}\,. \tag{7.1}$$

It is called "successor function" since it is the operation of adding (with the semiring addition $\oplus_{S,\beta}$) a copy of the multiplicative unit (which is 0 in the semiring multiplication $\odot$), just as the usual successor function of arithmetic $x \mapsto x+1$ is the operation of adding the multiplicative unit.

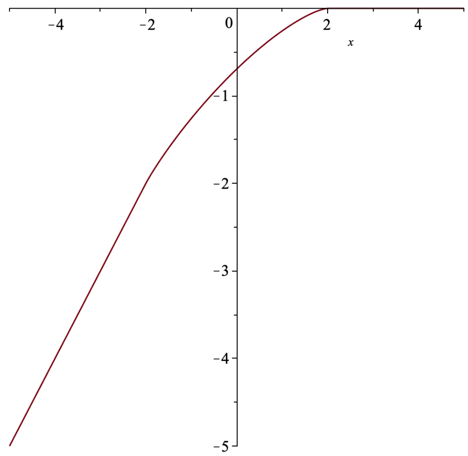


FIGURE 2. The behavior of the successor function $\Upsilon(x,\beta)$ for $S = \mathrm{Ry}_2$, shown in the case of $\beta = 1$.

The behavior in the case of $S = \mathrm{Ry}_2$ is shown in Figure 2. The property that $\Upsilon(x,\beta) \sim x$ for $x \to -\infty$ and $\Upsilon(x,\beta) \sim 0$ for $x \to +\infty$ is common to all thermodynamic semirings: these properties are in fact related to the existence of an additive unit in the semiring (Proposition 9.1 of [39]). In the specific case of $S = \mathrm{Ry}_2$, we obtain from the explicit form of the minimum $\lambda_{\min}(u)$ as in (3.10) (see Figure 1) that in fact $\Upsilon(x,\beta) = x$ for all $x < -2\beta^{-1}$ and $\Upsilon(x,\beta) = 0$ for all $x > 2\beta^{-1}$ and that in the interval $|x| < 2\beta^{-1}$ it is given by the function

$$\Upsilon(x,\beta) = -1 + u + \sqrt{1-u^2} + \beta^{-1} \log\left(\frac{1-\sqrt{1-u^2}}{u^2}\right) \tag{7.2}$$

for $u = \beta x/2$, with $|u| < 1$, which is shown in Figure 3. (While this is less evident in this plot than in the one of Figure 1, the function (7.2) is only piecewise smooth, with singularities at $u = \pm 1$.)

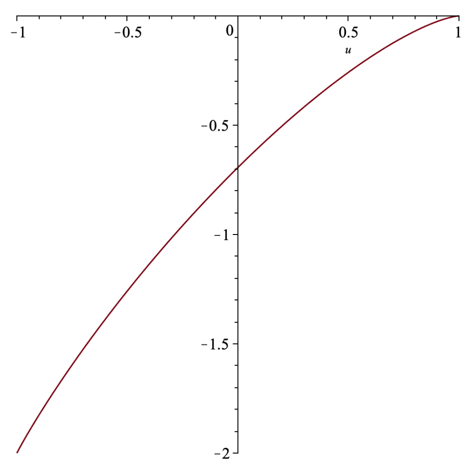


FIGURE 3. The successor function $\Upsilon(x,\beta)$ for $S = \mathrm{Ry}_2$ and $\beta = 1$, in the interval $|u| < 1$ for $u = \beta x/2$.

A direct computation from (7.2) with $u = \beta x/2$ gives the following property of the successor function, which will be useful in implementing Minimal Search.

Lemma 7.1. *The successor function $\Upsilon(x,\beta)$ with $S=\mathrm{Ry}_2$ has the following expansion in β*

$$\Upsilon(x,\beta)\sim -\log(2)\beta^{-1}+(\frac{1}{2}x+\frac{1}{16}x^2)\beta-\frac{1}{32}x^2\beta^2+\frac{3}{512}x^4\beta^3-\frac{1}{2048}x^4\beta^4+O(\beta^5)\,. \tag{7.3}$$

In particular the expansion does not contain any term of order zero in β.

Remark 7.2. The distributivity of the multiplication $\odot$ over the sum $\oplus_{S,\beta}$, in a thermodynamic semiring $\mathbb{T}_{S,\beta}$, namely the property

$$z\odot(x\oplus_{S,\beta}y)=z+\min_\lambda\{\lambda x+(1-\lambda)y-\beta^{-1}S(\lambda)\}$$
$$=\min_\lambda\{z+\lambda x+(1-\lambda)y-\beta^{-1}S(\lambda)\}=\min_\lambda\{\lambda(x+z)+(1-\lambda)(y+z)-\beta^{-1}S(\lambda)\}$$
$$=(x+z)\oplus_{S,\beta}(y+z)=(x\odot z)\oplus_{S,\beta}(y\odot z)\,,$$

implies that the addition $x\oplus_{S,\beta}y$ is completely determined by the successor function via

$$x\oplus_{S,\beta}y=\Upsilon(x-y,\beta)+y\,.$$

We can then provide a model of Merge as phase-synchronization of waves based on the construction of the previous sections and the properties of the successor function.

7.4. **Merge represented as phase synchronization.** Here we assume, as in the rest of this paper, that there is a representation

$$\varphi:\mathcal{SO}_0\to\mathcal{F}=\mathcal{C}^\infty(X,\mathbb{R})$$

of lexical items and syntactic features.

These functions $\varphi(\alpha)=\varphi_\alpha(t,x,y,z)$ can be thought of, in first approximation, as sinusoidal waves $\varphi_\alpha(t,x,y,z)=A_\alpha\sin(\nu_\alpha t+\omega_\alpha)$ where $A_\alpha=A_\alpha(x,y,z)$ is the amplitude, $\nu_\alpha=\nu_\alpha(x,y,z)$ if the frequency, and $\omega_\alpha=\omega_\alpha(x,y,z)$ is the phase. This assumption is not too unrealistic, as superpositions of few pure sinusoidal waves may in fact be adequate for word representation, [69].

We also do not discuss here the dependence of amplitude, frequency and phase on the spatial location: the variability may be encoded as a statistical model. This is adequate since the transversality arguments we use reply on a generic case (dense open set). We do assume, however, that the variation is sufficiently regular (at least continuous and piecewise smooth for the resulting representations of syntactic objects, and smooth for the waves φ_α representing lexical items and features).

7.4.1. *Phase synchronization.* Consider first the case of two waves with the same frequency $\nu_{\alpha_1}=\nu_{\alpha_2}$. In this case, phase-synchronization simply means that the phases are shifted so that the phase difference

$$\Delta\omega_{\alpha_1\alpha_2}=\omega_{\alpha_1}-\omega_{\alpha_2}$$

becomes zero. Such a phase shift could be achieved by an effect that moves both phases towards an average of the two. In view of the discussion on the algebraic properties of such averaging operations, one can instead consider an averaging that is entropy-optimized, as discussed above, so that ω_{α_1} and ω_{α_2} are replaced by a common synchronized phase

$$\omega_{\alpha_1}\oplus_{\mathrm{Ry}_2,\beta}\omega_{\alpha_2}\,. \tag{7.4}$$

Since in general one can evaluate *phase differences* rather than phases, one can equivalently write the phase synchronization (7.4) in terms of the successor function. We can write (7.4) as

$$\omega_{\alpha_1}\oplus_{\mathrm{Ry}_2,\beta}\omega_{\alpha_2}=\omega_{\alpha_2}+\Upsilon(\omega_{\alpha_1}-\omega_{\alpha_2},\beta)=\omega_{\alpha_2}\odot\Upsilon(\Delta\omega_{\alpha_1\alpha_2},\beta)\,, \tag{7.5}$$

which expresses the phase synchronization as applying the successor function of the thermodynamic semiring to the phase difference.

7.4.2. *Cross-frequency phase synchronization.* More generally, for different frequencies $\nu_{\alpha_1} \neq \nu_{\alpha_2}$, one can still achieve phase synchronization via the mechanism of *cross-frequency phase synchronization*. For this, we assume that the two frequencies are in a rational relation, namely there are relatively prime integers $n_{(\alpha_1,\alpha_2)}, n_{(\alpha_2,\alpha_1)} \in \mathbb{N}$ with $\gcd(n_{(\alpha_1,\alpha_2)}, n_{(\alpha_2,\alpha_1)}) = 1$, such that

$$n_{(\alpha_1,\alpha_2)}\nu_{\alpha_1} = n_{(\alpha_2,\alpha_1)}\nu_{\alpha_2}\,. \tag{7.6}$$

In this case the phase difference is measured as

$$\Delta\omega_{\alpha_1\alpha_2} := n_{(\alpha_1,\alpha_2)}\omega_{\alpha_1} - n_{(\alpha_2,\alpha_1)}\omega_{\alpha_2}\,. \tag{7.7}$$

We can then again synchronize the phases as above, with the synchronized phase

$$\begin{aligned} \omega_{\alpha_1}^{\odot n_{(\alpha_1,\alpha_2)}} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2}^{\odot n_{(\alpha_2,\alpha_1)}} &= n_{(\alpha_1,\alpha_2)}\omega_{\alpha_1} \oplus_{\mathrm{Ry}_2,\beta} n_{(\alpha_2,\alpha_1)}\omega_{\alpha_2} \\ &= \omega_{\alpha_2}^{\odot n_{(\alpha_2,\alpha_1)}} \odot \Upsilon(\Delta\omega_{(\alpha_1,\alpha_2)}, \beta)\,, \end{aligned} \tag{7.8}$$

with the phase difference $\Delta\omega_{\alpha_1\alpha_2}$ defined as in (7.7). In the case where the phases (and frequencies and amplitudes) of the sinusoidal waves have a dependence on the spatial coordinates (with locally constant n_{α_i}), the semiring operations are applied pointwise, as discussed in the previous sections.

To see the action of Merge as phase synchronization, assume that we start with a packet of wave functions

$$\Phi(t) = \sum_\alpha c_\alpha\, \varphi_\alpha(t)$$

with coefficients $c_\alpha \in \mathbb{R}_+$ (these can be absorbed into the amplitudes A_α), and with wave functions $\varphi_\alpha(t) = A_\alpha \sin(\nu_\alpha t + \omega_\alpha)$, where we leave implicit any possible dependence of A, ν, ω on the spatial coordinates.

The merging $\mathfrak{M}(\alpha_1, \alpha_2)$ then replaces a packet $\sum_\alpha \varphi_\alpha$ with a new packet in which the contribution $\varphi_{\alpha_1} + \varphi_{\alpha_2}$ of the two waves

$$\varphi_{\alpha_i} = A_{\alpha_i} \sin(\nu_{\alpha_i} t + \omega_{\alpha_i})$$

for $i = 1, 2$, is replaced, via cross-frequency phase synchronization, by the contribution of the phase shifted waves

$$\mathfrak{M}(\varphi_{\alpha_1}, \varphi_{\alpha_2}) = A_{\alpha_1} \sin(\nu_{\alpha_1} t + \omega_{\{\alpha_1,\alpha_2\}}) + A_{\alpha_2} \sin(\nu_{\alpha_2 i} t + \omega_{\{\alpha_1,\alpha_2\}}) \tag{7.9}$$

where

$$\omega_{\{\alpha_1,\alpha_2\}} := \omega_{\alpha_1}^{\odot n_{(\alpha_1,\alpha_2)}} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2}^{\odot n_{(\alpha_2,\alpha_1)}}$$

is obtained as above from the successor function of the thermodynamic semiring applied to the phase shift of (7.7).

More generally, one can assume that the φ_α are superpositions of a few sinusoidal waves $\varphi_\alpha = \sum_i A_{\alpha,i} \sin(\nu_{\alpha,i} t + \omega_{\alpha,i})$ and that the Merge operation gives a superposition of waves of the form $\varphi_{\mathfrak{M}(\alpha_1,\alpha_2)} = \sum_{ij}(A_{\alpha_1,i} \sin(\nu_{\alpha_1,i} t + \omega_{\{(\alpha_1,i),(\alpha_2,j)\}}) + A_{\alpha_1,j} \sin(\nu_{\alpha_1,j} t + \omega_{\{(\alpha_1,i),(\alpha_2,j)\}}))$ where the frequencies $\nu_{\alpha_1,i}$ and $\nu_{\alpha_1,j}$ have rational ratios. For simplicity of notation we just write the case where the φ_α are single sinusoidal waves, as this superposition case is analogous.

Proposition 7.3. *The operation* (7.9) *is commutative. It is also non-associative, for a dense open set of functions $\omega \in \mathcal{C}^\infty(\Omega, \mathbb{R}^N)$, with $\Omega \subset \mathbb{R}^3$ the spatial region where the waves are located, and $N = \#\mathcal{SO}_0$, such that $\varphi_\alpha = A_\alpha \sin(\nu_\alpha t + \omega_\alpha)$.*

Proof. The commutativity of the operation (7.9) follows from the fact that, while $n_{(\alpha_1,\alpha_2)} \neq n_{(\alpha_2,\alpha_1)}$, exchanging φ_{α_1} and φ_{α_2} exchanges both ω_{α_1} and ω_{α_2} as well as ν_{α_1} and ν_{α_2} hence it exchanges also $n_{(\alpha_1,\alpha_2)}$ and $n_{(\alpha_2,\alpha_1)}$, so that the result is the same, $\omega_{\alpha_1\alpha_2} = \omega_{\alpha_2\alpha_1}$, so we obtain as needed that

$$\mathfrak{M}(\varphi_{\alpha_1}, \varphi_{\alpha_2}) = \mathfrak{M}(\varphi_{\alpha_2}, \varphi_{\alpha_1})\,.$$

The non-associativity of the operation (7.9) lies in the nonassociativity of the iterated phase shifts

$$\omega_{\alpha_3}^{\odot n_{(\alpha_3,(\alpha_1,\alpha_2))}} \oplus_{\mathrm{Ry}_2,\beta} (\omega_{\alpha_1}^{\odot n_{(\alpha_1,\alpha_2)}} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2}^{\odot n_{(\alpha_2,\alpha_1)}})^{\odot n_{((\alpha_1,\alpha_2),\alpha_3)}} \neq$$
$$(\omega_{\alpha_3}^{\odot n_{(\alpha_3,\alpha_1)}} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_1}^{\odot n_{(\alpha_1,\alpha_3)}})^{\odot n_{((\alpha_3,\alpha_1),\alpha_2)}} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2}^{\odot n_{(\alpha_2,(\alpha_3,\alpha_1))}},$$

where the multiplicities are determined by comparing the frequencies

$$n_{((\alpha_1,\alpha_2),\alpha_3)} n_{(\alpha_1,\alpha_2)} \nu_{\alpha_1} = n_{((\alpha_1,\alpha_2),\alpha_3)} n_{(\alpha_2,\alpha_1)} \nu_{\alpha_2} = n_{(\alpha_3,(\alpha_1,\alpha_2))} \nu_{\alpha_3}$$

and

$$n_{((\alpha_3,\alpha_1),\alpha_2)} n_{(\alpha_3,\alpha_1)} \nu_3 = n_{((\alpha_3,\alpha_1),\alpha_2)} n_{(\alpha_1,\alpha_3)} \nu_1 = n_{(\alpha_2,(\alpha_3,\alpha_1))} \nu_2 .$$

This is again obtainable from a transversality argument as in Theorem 4.5, extended as in Theorem 4.6 to the case of iterations. In this case, the hypersurfaces where transversality needs to occur are of the form (for the case of Theorem 4.5)

$$H_{\underline{n}} = \{(x_1,x_2,x_3) \in \mathbb{R}^3 \,|\, x_3^{\odot u} \oplus_{\mathrm{Ry}_2,\beta} (x_1^{\odot a} \oplus_{\mathrm{Ry}_2,\beta} x_2^{\odot b})^{\odot v} = (x_3^{\odot c} \oplus_{\mathrm{Ry}_2,\beta} x_1^{\odot d})^{\odot p} \oplus_{\mathrm{Ry}_2,\beta} x_2^{\odot q}\} \tag{7.10}$$

with $\underline{n} = (n_{ij})_{i\neq j \in \{1,2,3\}}$ satisfying

$$\frac{n_{ij}}{n_{ji}} \frac{n_{jk}}{n_{kj}} = \frac{n_{ik}}{n_{ki}} \tag{7.11}$$

for $i \neq j \neq k$, and with $a = n_{12}$, $b = n_{21}$, $c = n_{31}$, $d = n_{13}$, and $u,v \in \mathbb{N}$ and $p,q \in \mathbb{N}$ are, respectively, pairs of relatively prime integers determined by the identities

$$\frac{v}{u} = \frac{n_{13}}{n_{31}n_{12}} = \frac{n_{23}}{n_{32}n_{21}} \quad \text{and} \quad \frac{p}{q} = \frac{n_{32}}{n_{23}n_{31}} = \frac{n_{12}}{n_{21}n_{13}} .$$

While there is a countable set of such $H_{\underline{n}}$, for varying choices of the integers in $\underline{n}$, in fact only a finite set of $H_{\underline{n}}$ is relevant and is determined by the set of frequencies $\{\nu_\alpha\}_{\alpha\in\mathcal{SO}_0}$ of the functions φ_α, through their ratios $\nu_j/\nu_i = n_{ij}/n_{ji}$. (In the case where we allow frequencies to vary with the spatial location in the compact region Ω, we assume their ratios are locally constant, hence in this case too only finitely many $\underline{n}$ are involved.) In the case of trees T, T' with $T \neq T'$ and with $L = L(T) = L(T')$ with the same assignment $\{\alpha_\ell\}_{\ell\in L}$ of $\alpha_\ell \in \mathcal{SO}_0$ at the leaves, we similarly have hypersurfaces $H_{T,T',\underline{n}}$ for $\underline{n} = (n_{ij})_{i\neq j\in\{1,\dots,n\}}$ for $n = \#L$, with the relations (7.11). These hypersurfaces extend the case of (7.10), by the appropriate modification of (4.11), using a version of $\mathcal{B}_{T,S,\beta,\underline{n}}(x_1,\dots,x_n)$ that incorporates the multiplicities n_{ij}. (We do not write these explicitly, but the construction is clear by iterating the expression in (7.10). Again, the number of hypersurfaces $H_{T,T',\underline{n}}$ that need to be taken into account for the transversality argument is finite, since the ratios n_{ij}/n_{ji} correspond to the rations of frequencies ν_j/ν_i in the finite set $\{\nu_\alpha\}_{\alpha\in\mathcal{SO}_0}$. □

In this model, the hierarchical construction of syntactic objects then proceeds in the following way. Given a finite set $\{\alpha_\ell\}_{\ell\in L}$ (assuming distinct $\alpha_\ell \neq \alpha_{\ell'}$ for $\ell \neq \ell'$), a syntactic object T with bracket notation $\mathcal{B}_T(\alpha_1,\dots,\alpha_n)$ is mapped to the function

$$\varphi_T := \sum_{\ell\in L} \omega_T(\varphi_{\alpha_\ell}) , \tag{7.12}$$

where ω_T here denotes the phase shift

$$\omega_T(\varphi_{\alpha_\ell})(t) = A_{\alpha_\ell} \sin(\nu_{\alpha_\ell} t + \omega_T)$$

where the synchronized phase ω_T takes the form

$$\omega_T = \mathcal{B}_{T,\mathrm{Ry}_2,\beta,\underline{n}}(\omega_{\alpha_1},\dots,\omega_{\alpha_n}) ,$$

for $\underline{n} = (n_{\ell\ell'})$ with $n_{\ell\ell'}/n_{\ell'\ell} = \nu_{\alpha_{\ell'}}/\nu_{\alpha_\ell}$.

7.5. **Accessible terms.** In addition to realizing the magma of syntactic objects, one needs to be able to identify accessible terms inside an already formed structure. In other words, one needs a Minimal Search algorithm for reconstructing the terms φ_{T_v} for $T_v \subset T$ an accessible term, given the wave function φ_T.

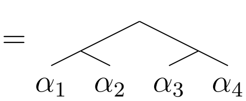
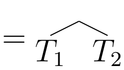
Since φ_T is obtained by repeated application of the semiring addition $\oplus_{S,\beta}$ to the phases, to implement cross-frequency phase synchronization, it is necessary to "undo" this operation to recover the original wave functions before synchronization. The semiring addition is not invertible, which seems to be a main obstacle to this operation. However, one can use some of the properties of the successor function analyzed above to extract the accessible terms.

We first consider the case of a syntactic object obtained from elements of $\mathcal{SO}_0$ via a single Merge operation, namely of the form $T = \mathfrak{M}(\alpha_1, \alpha_2)$. This syntactic object has only two accessible terms, α_1 and α_2. We need to show how to extract them from the wave function φ_T.

Lemma 7.4. *For a syntactic object $T \in \mathcal{SO}$ of the form $T = \mathfrak{M}(\alpha_1, \alpha_2)$ with $\alpha_1, \alpha_2 \in \mathcal{SO}_0$, it is possible to obtain the wave functions φ_{α_1} and φ_{α_2} from only observing the result of Merge φ_T, provided φ_T can be observed at variable values of the parameter β.*

Proof. We know by (7.12) that, in order to be able to access φ_{α_1} and φ_{α_2} we need to obtain the phases ω_{α_1} and ω_{α_2} from the synchronized phase $\omega_T = \omega_{\alpha_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2}$. We can assume without loss of generality that $\omega_{\alpha_2} > \omega_{\alpha_1}$, and we write $\omega_T = \omega_{\alpha_2} + \Upsilon(\omega_{\alpha_1} - \omega_{\alpha_2}, \beta)$, with $\omega_{\alpha_1} - \omega_{\alpha_2} < 0$. In the range $x \leq 1$ and for fixed β, the successor function $\Upsilon(x, \beta)$ is invertible. For $\xi = \Upsilon(x, \beta)$, we write $x = \Xi(\xi, \beta)$ for the inverse function. Thus, if we can extract the term ω_{α_2} from $\omega_T = \omega_{\alpha_2} + \Upsilon(\omega_{\alpha_1} - \omega_{\alpha_2}, \beta)$, then we can also obtain ω_{α_1} by inverting the successor function: $\omega_{\alpha_1} = \Xi(\Upsilon(\omega_{\alpha_1} - \omega_{\alpha_2}, \beta), \beta) + \omega_{\alpha_2}$. In order to extract the leading phase ω_{α_2} from ω_T, we use the property of Lemma 7.1, namely the fact that the expansion (7.3) in β of the successor function has no term of order zero in β, so that the only zero order term in β in ω_T is ω_{α_2}. In other words, if we consider β a tunable parameter, after removing the divergent polar part $-\log(2)\beta^{-1}$, for sufficiently small β all the terms in the expansion (7.3) with positive powers of β are negligible and one is left with the finite (renormalized) value ω_{α_2}. This shows that, for a single Merge operation, $T = \mathfrak{M}(\alpha_1, \alpha_2)$ the wave functions φ_{α_1} and φ_{α_2} representing the accessible terms are indeed accessible given the wave function φ_T. $\square$

The argument of Lemma 7.4 relies on the properties of the β-expansion of the successor function. It does not directly apply to hierarchical structures $T = \mathfrak{M}(T_1, T_2)$ with $\omega_T = \omega_{T_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T_2}$, where the $\omega_{T_1}, \omega_{T_2}$ themselves are iterations of previous $\oplus_{\mathrm{Ry}_2,\beta}$ operations, hence they incorporate a β-dependence. Indeed, it is difficult to iterate Lemma 7.4 to syntactic objects that involve multiple repeated applications of the magma operation $\mathfrak{M}$, because successive applications of $\mathfrak{M}$ compose the β-expansions, so that isolating a constant term no longer corresponds to isolating one of the two waves being merged. One can see this more explicitly by considering the example of a syntactic object of the form

(7.13) $T =$  $=$  $= \mathfrak{M}(T_1, T_2)$

where the corresponding compositions that implements the phase synchronization would be of the form

$$\omega_T = \omega_{T_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T_2}$$

with

$$\omega_{T_1} = \omega_{\alpha_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_2} \quad \text{and} \quad \omega_{T_2} = \omega_{\alpha_3} \oplus_{\mathrm{Ry}_2,\beta} \omega_{\alpha_4} \,.$$

In terms of the successor function, assuming $\omega_{T_2} > \omega_{T_1}$ and $\omega_{\alpha_2} > \omega_{\alpha_1}$ and $\omega_{\alpha_4} > \omega_{\alpha_3}$ (the other cases are analogous using commutativity of $\oplus_{\mathrm{Ry}_2,\beta}$, we have

$$\omega_T = \omega_{T_2} + \Upsilon(\omega_{T_1} - \omega_{T_2}, \beta) =$$
$$\omega_{\alpha_4} + \Upsilon(\omega_{\alpha_3} - \omega_{\alpha_4}, \beta) + \Upsilon\big(\omega_{\alpha_2} + \Upsilon(\omega_{\alpha_1} - \omega_{\alpha_2}, \beta) - \omega_{\alpha_4} - \Upsilon(\omega_{\alpha_3} - \omega_{\alpha_4}, \beta), \beta\big)\,.$$

Considering the β-expansions, we obtain

$$\begin{aligned}\omega_T \sim \omega_{\alpha_4} - 2\log(2)\beta^{-1} + f_1(\omega_{\alpha_3} - \omega_{\alpha_4})\beta \\ + f_1\big(\omega_{\alpha_2} - \omega_{\alpha_4} + (f_1(\omega_{\alpha_1} - \omega_{\alpha_2}) - f_1(\omega_{\alpha_3} - \omega_{\alpha_4}))\beta + O(\beta^2)\big)\beta + O(\beta^2) \\ \sim \omega_{\alpha_4} - 2\log(2)\beta^{-1} + \big(f_1(\omega_{\alpha_3} - \omega_{\alpha_4}) + f_1(\omega_{\alpha_2} - \omega_{\alpha_4})\big)\beta + O(\beta),\end{aligned}$$

where

$$f_1(x) = \frac{1}{2}x + \frac{1}{16}x^2\,.$$

This would allow us to extract ω_4 as the β-invariant term as in Lemma 7.4. However, here the β-expansion does not separate the terms $\Upsilon(\omega_{\alpha_3} - \omega_{\alpha_4}, \beta)$ and $\Upsilon\big(\omega_{\alpha_2} + \Upsilon(\omega_{\alpha_1} - \omega_{\alpha_2}, \beta) - \omega_{\alpha_4} - \Upsilon(\omega_{\alpha_3} - \omega_{\alpha_4}, \beta), \beta\big)$ to access the other ω_{α_i} through inversion of Υ. The following terms of the expansion provide polynomial expressions in the ω_{α_i}. Thus, a possible way to determine the ω_{α_i} would then be to solve these polynomial constraints. More generally, one can pose the following question.

Question 7.5. Provide a Minimal Search algorithm that extracts a given ω_{T_v}, for any chosen accessible term $T_v \subset T$, from the synchronized phase ω_T, observed as a function of the variable parameter β.

Note that this problem looks like an inversion of power series problem, which can be seen as essentially a Hopf algebraic renormalization problem, see [23] and the discussion of the Lagrange inversion formula in [27] and its interpretation as antipode in the Faà di Bruno Hopf algebra, [33], and the relation between the Faà di Bruno Hopf algebra and the Connes–Kreimer Hopf algebra of rooted trees, in the context of Dyson–Schwinger equations, [22]. We will not discuss this problem further in this paper.

Remark 7.6. If we allow a multivariable version, where each application of $\oplus_{S,\beta}$ corresponding to each non-leaf node v of the tree T can carry an independent parameter β_v, then the argument of Lemma 7.4 would extend directly to the extraction of ω_{T_v} from ω_T, for any accessible term T_v. The more difficult Question 7.5 pertains to the case where a single β-parameter is used.

Given an algorithm as discussed above that can extract ω_{T_v} from ω_T for any accessible term $T_v \subset T$, we then have the Internal Merge operation that maps

$$\omega_T \mapsto \omega_{T_v} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T/T_v}\,.$$

7.6. **Remark on Merge and the successor function.** In this framework, each application of Merge is realized as an application of the successor function of the thermodynamic semiring, applied to the phases, to achieve synchronization. As we have discussed, the synchronization of phases is achieved by a averaging optimized by a Rényi information functional $S = \mathrm{Ry}_2$, in the form of the thermodynamic semiring addition $\oplus_{S,\beta}$. Equivalently, both the External Merge operation $\omega_T = \omega_{T_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T_2}$ for $T = \mathfrak{M}(T_1, T_2)$ and the Internal Merge operation $\omega_T \mapsto \omega_{T_v} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T/T_v}$ are expressible in terms of the *successor function of the semiring*.

Formal similarities between Internal Merge and the *successor function of arithmetic* have been observed (for instance in [12]). Indeed the von Neumann set theoretic construction of the integers and the successor function of arithmetic, where one defines $0 = \emptyset = \{\ \}$, $1 = \{\emptyset\}$, $2 = \{\emptyset, \{\emptyset\}\}$,

$3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ and so on with the successor function $n \mapsto n+1$ defined in this set-theoretic form as $\Upsilon : A_n \mapsto A_{n+1} = A_n \cup \{A_n\}$. Since the only information that finite sets carry is their cardinality, the von Neumann construction indeed identifies a set-theoretic operation that is fully equivalent to the step $n \mapsto n+1$ leading from cardinality n to cardinality $n+1$, namely the successor function of arithmetic. A formal similarity with Internal Merge can be seen using the bracket notation for full binary rooted trees and writing the simplest Internal Merge in the form $\{\alpha_1, \alpha_2\} \mapsto \{\alpha_1, \{\cancel{\alpha_1}, \alpha_2\}\}$, with $\cancel{\alpha_1}$ denoting the trace left by the cancellation of the deeper copy. While this is an interesting and suggestive analogy, these two operations have different algebraic properties, hence they cannot be directly compared.

On the other hand, the model discussed shows that there is a precise way in which *both* Internal Merge *and* External Merge behave like a successor function, not the one of arithmetic but an appropriate one through which the algebraic properties of the Merge operation can be realized.

7.7. **Workspaces and Merge action as synchronization.** Consider the set of workspaces $\mathfrak{F}_{\mathcal{SO}_0}$, namely binary rooted forests $F = \sqcup_a T_a$ whose components are syntactic objects in $\mathcal{SO} = \mathfrak{T}_{\mathcal{SO}_0}$, and the linear span $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ (with real coefficients). We know from [35] that Merge (in all its forms: the optimal External and Internal Merge and the optimality violating Sideward Merge) can be realized as a Hopf algebra Markov chain

$$\mathcal{K} = \sqcup \circ (\mathcal{B} \otimes \mathrm{id}) \circ \Pi^{(2)} \circ \Delta \tag{7.14}$$

on $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ endowed with the disjoint union product $\sqcup$ and the coproduct Δ that extracts forests of disjoint accessible terms via admissible cuts on workspaces.

In the model discussed in this section, where we realize the magma operation $\mathfrak{M}$ as a phase synchronization on sinusoidal waves, we can also realize the Merge operations in the same Hopf algebraic terms discussed in [35].

Proposition 7.7. *The span $\mathcal{W}$ of finite products of wave functions φ_T is endowed with a bialgebra structure, as the polynomial algebra in the φ_T with coproduct*

$$\Delta\varphi_F = \sum_{F_{\underline{v}}} \varphi_{T_{v_1}} \cdots \varphi_{T_{v_n}} \otimes \varphi_{F/F_{\underline{v}}}\,, \tag{7.15}$$

for $\underline{v} = (v_1, \ldots, v_n)$ with $F_{\underline{v}} = T_{v_1} \sqcup \cdots \sqcup T_{v_n}$ a forest of disjoint accessible terms. The map $\varphi : T \mapsto \varphi_T$ gives a bialgebra homomorphism $\varphi : \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) \to \mathcal{W}$, from workspaces to their representation as waves. The Merge action on workspaces is then implemented as the unique linear map that makes the diagram commute:

$$\begin{array}{ccc} \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) & \xrightarrow{\varphi} & \mathcal{W} \\ {\scriptstyle\mathcal{K}}\downarrow & & \downarrow{\scriptstyle\mathcal{K}_\varphi} \\ \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) & \xrightarrow{\varphi} & \mathcal{W} \end{array} \tag{7.16}$$

where $\mathcal{K}$ is the Hopf algebra Markov chain (7.14), *with $\Pi^{(2)}$ the projection on the terms of the coproduct with a forest with two components in the left-channel. The map $\mathcal{K}_\varphi$ is also of the same Markov chain form.*

Proof. The first step is identifying what should play the role of workspaces $F \in \mathfrak{F}_{\mathcal{SO}_0}$. We need to associate to $F = \sqcup_a T_a$ a function φ_F, built out of the sine waves φ_{T_a}. It is natural to expect that this mapping $F \mapsto \varphi_F$ should give a morphism of commutative algebras. This means that the target space should be a polynomial algebra in the functions φ_T for $T \in \mathcal{SO}$ and that $\varphi_F = \prod_a \varphi_{T_a}$. Thus, this compatibility with the commutative algebra structure of the space of workspaces $\mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ implies that we encode workspaces F as *products* of sinusoidal waves. Thus, we consider as the target space $\mathcal{W}$ where workspaces are represented the linear span of products of sinusoidal waves φ_T for $T \in \mathcal{SO}$. On this space $\mathcal{W}$, we consider a coproduct of the form (7.15).

The extraction of accessible terms from a product $\varphi_F = \prod_a \varphi_{T_a}$ is possible, given an algorithm for the extraction of the phases ω_{T_v} of the accessible terms T_v of the individual components T_a from the phase ω_{T_a} of the wave function φ_{T_a}, as discussed in §7.5. Indeed, it suffices to show that we can recover the phases ω_{T_a} from the product φ_F, since then all the phases ω_{T_v} would also be recoverable, while the frequencies and amplitudes are determined by those of the φ_α for the terms α at the leaves of F. All the φ_{T_v} are then recoverable, so that the extraction of accessible terms in (7.15) is fully computable. For sine waves $\varphi_{T_a} = \sum_i A_{T_{a,i}} \sin(\nu_{a,a} t + \omega_{T_a})$, the phases ω_{T_a} are recoverable from the finite product $\varphi_F = \prod_a \varphi_{T_a}$ via Fourier transform, which for a product of sine waves is given by a combination of delta functions from whose weights the phases ω_{T_a} can be directly obtained (we can assume that the φ_{T_a} all have distinct frequencies ν_a). Thus, the map $\varphi : \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) \to \mathcal{W}$ then gives a bialgebra homomorphism.

Given the Hopf algebra Markov chain $\mathcal{K} : \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0}) \to \mathcal{V}(\mathfrak{F}_{\mathcal{SO}_0})$ of the Merge action on workspaces (7.14), there is a corresponding $\mathcal{K}_\varphi : \mathcal{W} \to \mathcal{W}$ that makes the diagram (7.16) commute. The map $\mathcal{K}_\varphi : \mathcal{W} \to \mathcal{W}$ takes the form

$$\mathcal{K}_\varphi = \mu \circ (\mathbb{B} \otimes \mathrm{id}) \circ \Pi^{(2)} \circ \Delta \,,$$

for Δ the coproduct in (7.15) and μ the multiplication in the algebra $\mathcal{W}$, and with

$$\mathbb{B}(\varphi_{T_1} \cdot \varphi_{T_2}) = \varphi_{\mathfrak{M}(T_1,T_2)} \,,$$

where $\varphi_{\mathfrak{M}(T_1,T_2)}$ is computed from the product $\varphi_{T_1} \cdot \varphi_{T_2}$ by first extracting the phases ω_{T_1} and ω_{T_2}, then computing from them $\omega_{\mathfrak{M}(T_1,T_2)} = \omega_{T_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T_2}$ and from this the resulting $\varphi_{\mathfrak{M}(T_1,T_2)}$. $\square$

7.8. **Further question: introducing head functions.** We have mentioned head functions on syntactic objects in §2.1, where a head function provided us with a choice at each internal node v of either a factor λ_v or $1 - \lambda_v$, so that the resulting probability distribution $A = (a_\ell)_{\ell \in L(T)}$ on the leaves of the tree depends not just on T, but on the pair (T, h_T). This dependence, however, is rather mild, and it eventually disappears entirely when the optimization

$$\min_{\lambda_v} \{\lambda_v x_v + (1 - \lambda_v) y_v - \beta^{-1} S(\lambda_v)\}$$

is computed, because of the symmetry $S(\lambda_v) = S(1 - \lambda_v)$.

On the other hand, the presence of a head function is important, both for the phase structure in syntax and the labeling algorithm, and for parsing in semantics. On the other hand, we know (§1.13 of [35] and [37]) that $\mathrm{Dom}(h) \subset \mathcal{SO}$ is not a submagma. For $T, T' \in \mathrm{Dom}(h)$, the failure of $\mathfrak{M}(T, T')$ being in $\mathrm{Dom}(h)$ is coming from exocentric constructions $\{\mathrm{XP}, \mathrm{YP}\}$ where there is no a priori indication of which should be the head. The hypermagma structure discussed in [37] accounts for this issue.

The operation $\oplus_{\mathrm{Ry}_2,\beta}$ wipes out any explicit information on the head function encoded in the probability distribution $A = (a_\ell)_{\ell \in L(T)}$ (hence making it possible to encode the magma structure), so the presence of the head has to be encoded here in a different way that is not directly involved in the phase-to-phase synchronization where $\oplus_{\mathrm{Ry}_2,\beta}$ is acting.

For example, the model of [54], [56] proposes phase-amplitude coupling (PAC) as a possible realization of headedness, where the phase of a lower frequency wave modulates the amplitude of a higher frequency wave, see Figure 4.

In the setting we are describing here, this would mean that when two waves φ_{T_1} and φ_{T_2} are merged in to the resulting wave $\varphi_{\mathfrak{M}(T_1,T_2)}$ according to the phase synchronization $\omega_{\mathfrak{M}(T_1,T_2)} = \omega_{T_1} \oplus_{\mathrm{Ry}_2,\beta} \omega_{T_2}$, as discussed above, if the head function $h_{\mathfrak{M}(T_1,T_2)}$ assigns to the root vertex v of

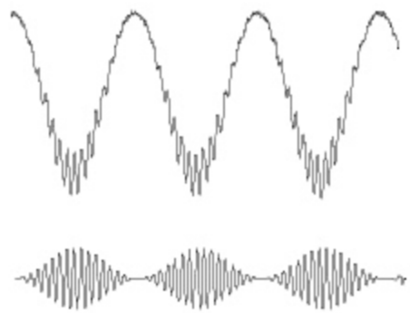


FIGURE 4. Phase-amplitude coupling in sinusoidal waves.

$\mathfrak{M}(T_1, T_2)$ the same head $h_{\mathfrak{M}(T_1,T_2)}(v) = h_{T_1}(v_1)$, where v_1 is the root vertex of T_1, then a lower frequency wave modulates the amplitude of the wave $\omega_T(\varphi_{T_1})$ in the resulting superposition wave $\varphi_{\mathfrak{M}(T_1,T_2)} = \omega_T(\varphi_{T_1}) + \omega_T(\varphi_{T_2})$.

Implementing this idea, however, has the problem that it leaves open the question of how the slower wave modulating the faster wave $\omega_T(\varphi_{T_1})$ that carries the head is to be determined: a mechanism for selecting such a modulating wave is not part of the type of model we have been discussing in this paper.

An alternative approach to incorporating head functions may be based on the results of [36] and [37], where the filtering of syntactic objects by theta roles assignments and by head, complement, specifier structure and well formed phases can be formulated in terms of a colored operad and its generating system. This suggests that in a representation of syntactic objects in terms of superposition of phase-synchronized waves, the different local coloring rules that mark head, complement, specifier, and modifier positions in the phase structure, and theta role assignments, may possibly correspond to different PAC modulations.

At present, the proposed theoretical models of Merge implementation available in the literature on neuroscience of language are not yet in a form that can be directly translated into a full mathematical formalization. Thus, for the moment, we have just provided some general possible models of mathematical operations that can be realized in a space of wavelets and that we can show faithfully capture the same algebraic properties of Merge.

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